

MAGNETOM

# Configuration System

IT Integration Instructions syngo MR XA

## Document Version

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## **1 Purpose of the Document**

This document provides information about the configuration and integration of the syngo Native based MAGNETOM system.

It is not intended for updating the software.

It contains descriptions of the tasks to be performed in sequence to achieve a system ready for use by a customer.

Besides the technical task descriptions, this document gives hints and recommendations for configuration. Be aware, that these are highly dependent on the customer situation / preference and need to be closely aligned with the MTRA/ Radiologist / IT Interfaces

## 2 Abbreviations

|           |                                                     |
|-----------|-----------------------------------------------------|
| AET       | Application Entity Title                            |
| AV object | Advanced application object                         |
| DICOM     | Digital Imaging and Communications in Medicine      |
| DIS       | Departmental Information System (e.g. RIS)          |
| DMWL      | DICOM Modality Worklist                             |
| DNS       | Domain Name Server                                  |
| eMR       | enhanced MR DICOM                                   |
| eMRC      | Enhanced MR Color                                   |
| FA        | Fractional Anisotropy                               |
| GSPS      | Grayscale Softcopy Presentation State               |
| HIS       | Hospital information System                         |
| IDEA      | Integrated Development Environment for Applications |
| LUT       | Look up Table                                       |
| MF C SC   | Multiframe Color Secondary Capture                  |
| MF SC     | Multiframe Grayscale Word Secondary Capture         |
| MPPS      | Modality Performed Procedure Steps                  |
| MWL       | Modality Worklist                                   |
| PACS      | Picture Archiving and Communication System          |
| RIS       | Radiology Information System                        |
| PR        |                                                     |
| SC        | Secondary Capture                                   |
| SCP       | Service Class Provider                              |
| SCU       | Service Class User                                  |
| SOP       | Service-Object Pair                                 |
| UID       | Unique Identifier                                   |
| UPS       | Unified Procedure Step                              |
| VF        | Ventricular Function                                |
| WINS      | Windows Internet Name Service                       |

|                                                                                     |                                                                  |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------|
|  | Configuration icon. Click on this to further configure a service |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------|

## 3 Quick Guide

### 3.1 Configure new PACS

**Hint:**

When creating a new Remote node, all Services should be included initially. It is not possible to remove or add a service after the node is saved. The Node needs be deleted and newly configured.

To configure a new PACS the following steps /chapters should be completed:

- Add new Remote DICOM Node (Chapter 5.2.3)
- Configure the following Services:
  - Storage (Chapter 5.2.5)
  - Query / Retrieve (Chapter 5.2.6)
  - Storage Commitment (Chapter 5.2.7)
- Configure Archiving Settings (Chapter 6.2.2)
- Configure Autorouting Rules (Chapter 6.4)

### 3.2 Configure new RIS

**Hint:**

When creating a new Remote node, all Services should be included initially. It is not possible to remove or add a service after the node is saved. The Node needs be deleted and newly configured.

To configure a new RIS the following steps /chapters should be completed:

- Add new Remote DICOM Node (Chapter 5.2.3)
- Configure the following Services:
  - MPPS (Chapter 5.2.8)
  - Worklist
- Configure Worklist Settings in Local Node (Chapter 5.1.3)
- Configure MPPS Completion (Chapter 6.6)

### 3.3 Configure Burn Robots

**Note:**

As it cannot be guaranteed that the referring physician is able to read Enhanced MR DICOM files and Burn Robots usually does not provide the functionality to convert Enhanced images.

***Recommendation: always configure Burn Robots in Interoperability mode.***

If the customer decides to use the Enhanced node here, then disable all MR Export rules.

To configure Burn Robots configure the following steps /chapters should be completed:

- Add New DICOM Role
  - Role: "Customized"
  - Manufacturer: "Other"
- Enter Hostname, IP Address, Logical Name, AET and PORT
- Press "Unlock"
- Delete all AET and Port information of all services besides Storage
- **Configure "Storage" for full interoperability (see 5.2.5 for details)**
- **Save**

## 3.4 Configure a teamplay Node

### Note:

- While teamplay Images and Receiver are capable to fully handle enhanced DICOM images, the applications, that get the images via teamplay may not. *Recommendation: configure the DICOM Node for Interoperability.*
- When scanning Bold sequences usually a large number of frames are created per multiframe image. Converting these from enhanced to interoperability results in a high number of individual images, since every frame becomes an individual image. This increases network traffic, may impact performance, and may create problems on applications / PACS systems. *Recommendation: keep Bold images as enhanced MR images.*
- teamplay does not support storage commitment
- The settings of the “Multiframe Conversion Result Structure” are purely a customer preference and should be discussed with the customer.

To configure a teamplay DICOM Node the following steps /chapters should be completed:

- Add New DICOM Role
  - Role: “teamplay”
- Enter Hostname, IP Address, Logical Name, AET and PORT
  - Only 2 services are available: Storage SCP and Query/Retrieve SCUDepending if teamplay images or a teamplay receiver are to be configured the default settings are as follows: Please confirm these settings with the customer as they may have been changed.
  - teamplay Images default configuration:
    - AE title: TEAMPLAY-IShare
    - Host name: formaltestREC
    - Port: 105
  - teamplay receiver default configuration (aka. teamplay analytics):
    - AE title: TEAMPLAY
    - Host name: formaltestREC
    - Port: 104
- Press Save
- **Configure “Storage”**
  - **Disable Storage Commitment**
  - **MR Export Rules for full interoperability (see 5.2.5 for details)**
- Press Save

MR Export Rules

MR Image Conversion

- ☒ Convert GSPS for MR Images

Multiframe Conversion

- ☒ Convert Enhanced MR and related GSPS
- ☒ Exclude BOLD Images
- ☒ Convert Enhanced MR Color LUT and related GSPS
- ☒ Convert Enhanced MR Color
- ☒ Convert Multiframe Color SC
- ☒ Convert Multiframe Grayscale Word SC and related GSPS

Do not convert Enhanced MR Images for remote syngo.via nodes because of interoperability reasons

Multiframe Conversion Result Structure

- ☒ Preserve original Series Number for each converted Multiframe Image
- ☐ Separate Series for each converted Diffusion Multiframe Image
- ☒ Separate Series for each converted Multiframe Image (all other than Diffusion, e.g. Twist, Grasp, ...)

OK Cancel



## 4 User Roles and Administration

### Initialize the clinadmin

Whilst the user clinadmin is preconfigured, no default password is set for the customer. This needs to be done as part of the Start-up.

- Login as Other User
  - Username: clinadmin
  - Password: (admin password - contact Siemens Service for support)
- Set new password as requested by customer
  - The password is case-sensitive.
  - Make sure, that the new password contains at least **14 characters**.  
Use **capital letters**, **lower case letters**, **numbers**, and **special characters**.

### Note:

- Lock out policy: after 10 wrong Password entries the account will be locked.
- When the customer has an IDEA license a new User has to be configured and be given the Role "MR\_IDEA".
- Never change role or task names!
- Only user accounts that are assigned to at least one *syngo* role are able to log on.
- Refer to syngo MR XA Operator Manual – Advance User for further information

The following Users are configured exFactory with the respective Roles:

The role "Technologist" is mapped to "Everyone"

| User                    | Role                   | For clinical use |
|-------------------------|------------------------|------------------|
| Meduser                 | Technologist           | x                |
| Medclient<br>(for MRWP) | Technologist           | x                |
| clinadmin               | Clinical Administrator | x                |
| MedAdmin                | IT Administrator       | -                |
| (No default user)       | MR_Idea                | (Option)         |
| (No default user)       | MR_Service             | -                |

The following rights are connected to the roles ("-" not connected; "x" connected):

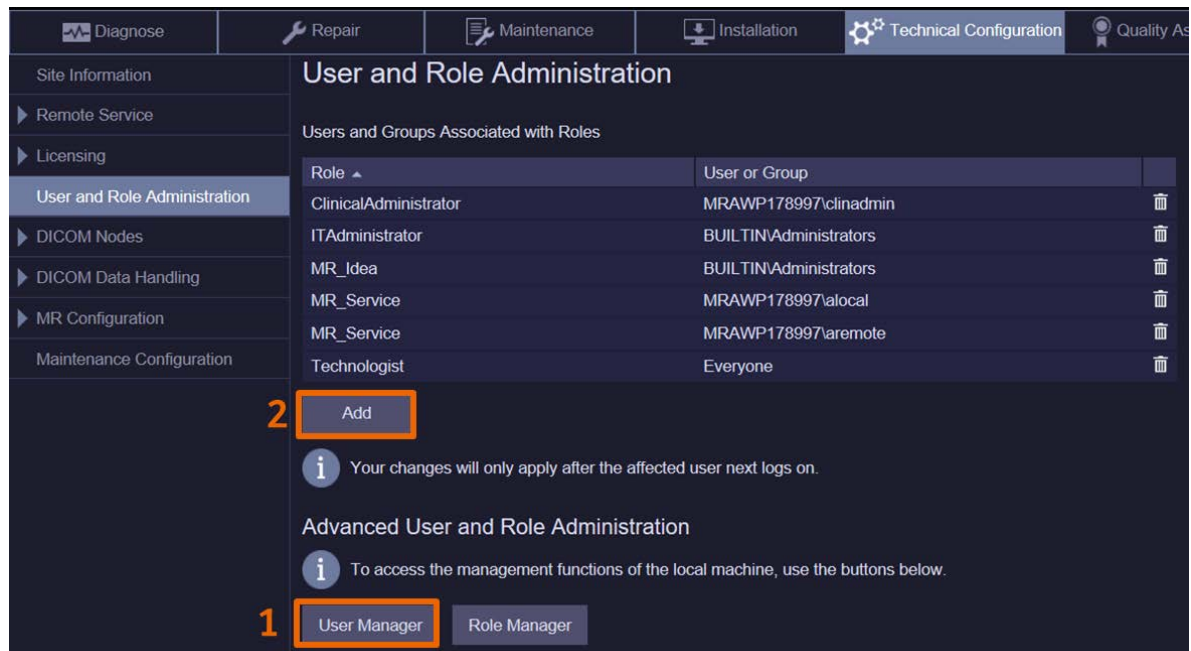
| Task name                                 | Technologist | Clin Admin | IT Admin | Researcher | MR_Idea | MR_Service | Authorized function                                                                                                                                                                                               |
|-------------------------------------------|--------------|------------|----------|------------|---------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AccessControl_Require_Authentication      | -            | -          | x        | -          | -       | -          | Enable or disable explicit authentication in the Configuration Panel.                                                                                                                                             |
| AdvancedLayoutOperations                  | -            | x          | x        | -          | -       | -          | Perform the following operations with public layouts in the Layout Gallery: <ul style="list-style-type: none"> <li>• switch to a layout</li> <li>• set a layout as the default</li> <li>• sort layouts</li> </ul> |
| COMPLETE_READ_WORKITEM                    | x            | -          | -        | -          | -       | -          | Save a workflow and to send the results to the archive.                                                                                                                                                           |
| CONF_WORKPLACE_SETTINGS – PERMISSION      | -            | x          | x        | -          | -       | -          | Configure the idle time after which the workplace is locked and a lock screen is activated to protect the system against unauthorized access.                                                                     |
| DATA_CORRECTION_AND_REARRANGEMENT         | x            | x          | x        | -          | -       | -          | Allows correction and rearrangement of patient and study data.                                                                                                                                                    |
| DATA_PERMANENT_DELETION                   | x            | x          | x        | -          | -       | -          | Allows data to be permanently deleted from the local database.                                                                                                                                                    |
| FavoritesToolbox.Site.Edit                | -            | x          | x        | -          | -       | -          | Customize the content of the Favorite Tools area for all users.                                                                                                                                                   |
| FavoritesToolbox.User.Edit                | x            | -          | -        | -          | -       | -          | Customize the content of the Favorite Tools area for the current user.                                                                                                                                            |
| INSTALL_SOFTWARE                          | -            | -          | x        | -          | -       | -          | Allows installation of software updates and upgrades provided by Siemens.                                                                                                                                         |
| MR_DC-ChangeTreePassword Without-Password | -            | x          | x        | -          | -       | x          | Change a tree password without knowing the password.                                                                                                                                                              |
| MR_ImpExpRetroReconAnd PhysioData         | x            | x          | x        | -          | x       | x          | See all performed image reconstructions in the Job Viewer, to store their raw data, and to start retrospective reconstruction with this data.                                                                     |

| Task name                          | Technologist | Clin Admin | IT Admin | Researcher | MR_Idea | MR_Service | Authorized function                                                                                                                                                                                                            |
|------------------------------------|--------------|------------|----------|------------|---------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MR_LeaveKioskMode                  | -            | -          | -        | x          | -       | -          | Deactivate kiosk mode without being prompted for credentials. This task should only be assigned to specific users, <b>never</b> to the group "Everyone". Otherwise anyone with access to the system may deactivate kiosk mode. |
| MR_SequenceDevelopment Watchdog    | -            | -          | -        | -          | x       | x          | Restart or reboot the MARS within the sequence development workflow.                                                                                                                                                           |
| NAV_ADMINISTRATION_MODE            | x            | x          | x        | -          | -       | -          | Allows access to the following functions of the Patient Browser: <ul style="list-style-type: none"> <li>• display of internal DICOM objects</li> <li>• display of TaskflowID in the result list</li> </ul>                     |
| NAV_ARCHIVE_STATE_CHANGE           | x            | x          | x        | -          | -       | -          | Prevents marking or allows marking of data for automatic archiving.                                                                                                                                                            |
| NAV_COMMUNICATION_EXPORT           | x            | x          | x        | -          | -       | -          | Allows export of data from the local database to a remote node or to removable media.                                                                                                                                          |
| NAV_COMMUNICATION_IMPORT           | x            | x          | x        | -          | -       | -          | Allows import of data from the file system to the local database in the Patient Browser.                                                                                                                                       |
| NAV_COMMUNICATION_SEND_TO_ARCHIVE  | x            | x          | x        | -          | -       | -          | Directly send data from the local database to the configured default archive in the Patient Browser.                                                                                                                           |
| NAV_EXTERNAL_DICOM_ACCESS          | x            | x          | x        | -          | -       | -          | Allows access to DICOM Query/Retrieve from the Patient Browser.                                                                                                                                                                |
| NAV_PUBLIC_FILTER_MANIPULATION     | x            | x          | x        | -          | -       | -          | Allows creation, modification, and deletion of public work lists.                                                                                                                                                              |
| NAV_THIRD_PARTY_APPLICATION_CALLUP | -            | -          | -        | -          | -       | -          | Start third-party applications from within the Patient Browser (via icon and context menu).                                                                                                                                    |

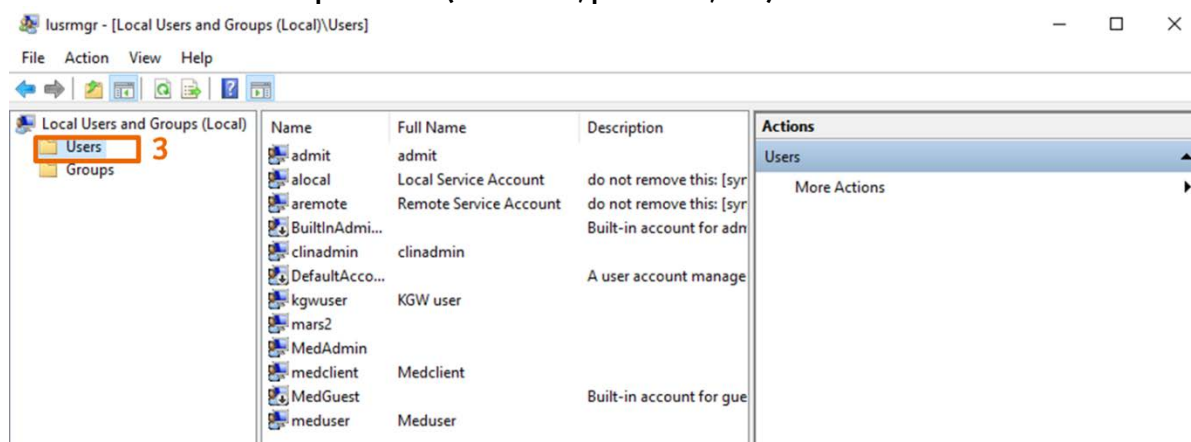
| Task name                               | Technologist | Clin Admin | IT Admin | Researcher | MR_Idea | MR_Service | Authorized function                                                                                                                                                                       |
|-----------------------------------------|--------------|------------|----------|------------|---------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NAV_WORKFLOW_ADMINISTRATION             | x            | x          | x        | -          | -       | -          | Allows access to the following workflow administration functions in the Patient Browser: <ul style="list-style-type: none"> <li>cancel workflow</li> <li>batch assign workflow</li> </ul> |
| PERFORM_CLINICAL_ADMIN_TASK             | -            | x          | x        | -          | -       | -          | Allows configuration of clinical settings in the syngo .via Administration Portal (e.g. configuration of anatomy processing).                                                             |
| PERFORM_IT_ADMIN_TASK                   | -            | -          | x        | -          | -       | -          | Allows configuration of IT settings in the syngo .via Administration Portal (e.g. license configuration).                                                                                 |
| PresetShortcut.Site.Edit                | -            | x          | x        | -          | -       | -          | Modify the keyboard shortcuts for windowing presets for all users.                                                                                                                        |
| PresetShortcut.Site.Restore             | -            | x          | x        | -          | -       | -          | Restore the factory settings of the keyboard shortcuts for windowing presets for all users.                                                                                               |
| PresetShortcut.User.Edit                | x            | -          | -        | -          | -       | -          | Modify the keyboard shortcuts for windowing presets for the current user.                                                                                                                 |
| PresetShortcut.User.Restore             | x            | -          | -        | -          | -       | -          | Restore the factory settings of the keyboard shortcuts for windowing presets for the current user.                                                                                        |
| Print.ReArrangeLayouts                  | -            | x          | x        | -          | -       | -          | Rearrange layouts within the Layout Gallery of the Print step.                                                                                                                            |
| PRIOR_QR_SOURCE_AND_FILTER_MANIPULATION | x            | x          | x        | -          | -       | -          | Modify the data sources and filter conditions in the Add Study dialog.                                                                                                                    |
| SaveDefaultLayoutForAllUsers            | -            | x          | x        | -          | -       | x          | Configure whether all clinical users can create and save private layout collections.                                                                                                      |
| SEE_HIDDEN_JOBS                         | -            | x          | -        | -          | -       | -          | See jobs that are marked as hidden in the Job View (e.g. DICOM communication jobs).                                                                                                       |
| TROUBLESHOOT_SYSTEM                     | -            | x          | x        | -          | -       | -          | Grants access to functions for error analysis and troubleshooting (e.g. Message Viewer).                                                                                                  |
| VrtPreset.Site.Create                   | -            | x          | x        | -          | -       | -          | Create VRT presets for all users.                                                                                                                                                         |
| VrtPreset.Site.Edit                     | -            | x          | x        | -          | -       | -          | Modify VRT presets for all users.                                                                                                                                                         |

| <b>Task name</b>             | <b>Technologist</b> | <b>Clin Admin</b> | <b>IT Admin</b> | <b>Researcher</b> | <b>MR_Idea</b> | <b>MR_Service</b> | <b>Authorized function</b>                                              |
|------------------------------|---------------------|-------------------|-----------------|-------------------|----------------|-------------------|-------------------------------------------------------------------------|
| VrtPreset.Site.Restore       |                     | x                 | x               | -                 | -              | -                 | Restore factory settings of VRT presets for all users.                  |
| VrtPreset.User.Create        | x                   | -                 | -               | -                 | -              | -                 | Create VRT presets for the current user.                                |
| VrtPreset.User.Edit          | x                   | -                 | -               | -                 | -              | -                 | Modify VRT presets for the current user.                                |
| VrtPreset.User.Restore       | x                   | -                 | -               | -                 | -              | -                 | Restore the factory settings of VRT presets for the current user.       |
| WindowingPreset.Site.Create  | -                   | x                 | x               | -                 | -              | -                 | Create windowing presets for all users                                  |
| WindowingPreset.Site.Edit    | -                   | x                 | x               | -                 | -              | -                 | Modify windowing presets for all users.                                 |
| WindowingPreset.Site.Restore | -                   | x                 | x               | -                 | -              | -                 | Restore factory settings of windowing presets for all users.            |
| WindowingPreset.User.Create  | x                   | -                 | -               | -                 | -              | -                 | Create windowing presets for the current user.                          |
| WindowingPreset.User.Edit    | x                   | -                 | -               | -                 | -              | -                 | Modify windowing presets for the current user.                          |
| WindowingPreset.User.Restore | x                   | -                 | -               | -                 | -              | -                 | Restore the factory settings of windowing presets for the current user. |

## 4.1 Adding a new User



- Navigate to *Technical Configuration* → *User and Role Administration*
- Click on [1] *User Manager*
- Enter Username: medadmin
- Enter password: (contact Siemens Service for support)
- Click [3] *Users*
- Navigate to *Action* → *New User...*
- Enter new user information (Username, password, etc.)



## 4.2 Linking a Role to a User

### Note:

The role "Technologist" is mapped to "Everyone".

- Navigate to *Technical Configuration* → *User and Role Administration*
- Click on [2] Add
- A new Row appears
- Choose Role from Dropdown [4]
- Enter User [5] (domain \ username)  
by default the domain equals that of the standard user: clinadmin, that can be found in the existing *User and Groups Associated with Roles* list
- Press save [6]

**User and Role Administration**

Users and Groups Associated with Roles

| Role ▲                | User or Group          |    |
|-----------------------|------------------------|----|
| ClinicalAdministrator | MRAWP178997\clinadmin  | 🗑️ |
| ITAdministrator       | BUILTIN\Administrators | 🗑️ |
| MR_Idea               | BUILTIN\Administrators | 🗑️ |
| MR_Service            | MRAWP178997\alocal     | 🗑️ |
| MR_Service            | MRAWP178997\aremote    | 🗑️ |
| Technologist          | Everyone               | 🗑️ |

Role Windows User or Group (e.g., domain\accountname) 📁 ✕

**4** **5** **6**

**i** Your changes will only apply after the affected user next logs on.

**Advanced User and Role Administration**

**i** To access the management functions of the local machine, use the buttons below.

User Manager Role Manager

### Hint:

to give a user the rights to enter the User Manager and Role Manager, the user also has to be given Administrator rights in the User Manager.

## 5 DICOM Nodes

The DICOM Nodes can be configured using the following path  
Administration Portal → Technical Configuration → DICOM Nodes

### 5.1 Local DICOM Node

In Local DICOM Node the MR System side can be configured.  
The following Items need to be configured:

- Worklist [2] see 5.1.3 for details
- Print settings [3] see 5.1.4 for details
- General settings [4] see 5.1.5 for details

#### 5.1.1 Prerequisites

- Which RIS is the Worklist provider (DIS)
- RIS Query interval
- Frequency of the DICOM Modality Worklist (DMWL) query in minutes.
- Time to keep unassigned Worklist items after disappearance from DIS
- Time window for scheduled procedure steps start date (days into the past )
- Time window for scheduled procedure steps start date (days into the future)



## 5.1.2 Local DICOM settings

By default, the AE title of the system is set to the hostname in uppercase and the port number is 104. The configuration can be changed in the administration portal **Technical Configuration > DICOM Nodes > Local DICOM Node**.

### Warning:

If the Local AE Title is deleted (press x and save) then the complete DICOM communication is lost! Contact Siemens Service / RSC for support.

### Hint:

Changes to the AE Title will be transferred to the different services as soon as **Save** is pressed.

Local DICOM Node

Role:  Modality:  Manufacturer:  Siemens Healthcare Model:  MR MAGNETOM Vida

Host Name:  MRAWP178997  
IP Address:   
Logical Name:  MRAWP178997  
Location:

| AE Title    | Port | Service           |
|-------------|------|-------------------|
| MRAWP178997 | 104  | Storage           |
| MRAWP178997 | SCU  |                   |
| MRAWP178997 | 104  | Query/Retrieve    |
| MRAWP178997 | SCU  |                   |
| MRAWP178997 | 104  | StorageCommitment |
| MRAWP178997 | SCU  |                   |
| MRAWP178997 | SCU  | Mpps              |
| MRAWP178997 | SCU  | Worklist          |
| MRAWP178997 | SCU  | Print             |

General Settings

Save

We recommend keeping the default settings unless the customer requests otherwise. The following restrictions must be considered when changing the AE title:

- Maximum of 16 alphanumeric characters
- White spaces and double quotes are not allowed
- Using capital letters is recommended
- Currently only one AET can be configured

### 5.1.3 Worklist Settings

#### Hint:

- Before configuring the Worklist settings, the RIS needs to be configured as a remote DICOM node in the administration portal. This is required to set the Worklist provider (DIS).
- When the customer is editing the patient information while a Query is updating the Worklist, the customer may drop out of the editing mode, losing the changes. Due to this the Query interval should not be set below 15 minutes.

## Local DICOM Node

### Worklist Query Settings

|   |                                                                     |          |                      |
|---|---------------------------------------------------------------------|----------|----------------------|
| 1 | Worklist provider (DIS)                                             | DVTK_RIS |                      |
| 2 | Query interval                                                      | 15       | minutes (off if 0)   |
| 3 | Time to keep unassigned worklist items after disappearance from DIS | 30       | minutes              |
|   | Time window for scheduled procedure step start date                 | 1        | days into the past   |
|   | and                                                                 | 1        | days into the future |

### Worklist Query Attributes and their Values

Select return key attributes for Worklist Query

- ✓ (0010,0010) Patient's Name
- ✓ (0010,0020) Patient ID
- ✓ (0010,0021) Issuer of Patient ID
- ✓ (0010,0030) Patient's Birth Date
- ✓ (0010,0032) Patient's Birth Time
- ✓ (0010,0040) Patient's Sex
- ✓ (0010,0050) Patient's Insurance Plan Code Sequence
- ✓ (0010,1000) Other Patient IDs

### Choose matching criteria to use for Worklist Query

|         | (0008,0060)<br>Modality | (0040,0001)<br>Application<br>Entity Title | (0040,0006)<br>Performing<br>Physician | Select Attribute | Select Attribute |
|---------|-------------------------|--------------------------------------------|----------------------------------------|------------------|------------------|
| Query 1 | MR                      | VIDA_AET                                   |                                        |                  |                  |
| Query 2 |                         |                                            |                                        |                  |                  |
| Query 3 |                         |                                            |                                        |                  |                  |
| Query 4 |                         |                                            |                                        |                  |                  |
| Query 5 |                         |                                            |                                        |                  |                  |
| Query 6 |                         |                                            |                                        |                  |                  |
| Query 7 |                         |                                            |                                        |                  |                  |
| Query 8 |                         |                                            |                                        |                  |                  |

- [1]From the **Worklist provider (DIS)** list, select the **RIS** which will be queried.
- [2]In the **Query interval** field, define the frequency of the DICOM Modality Worklist query in minutes. (*Recommended interval: >15 minutes*)
- [3]In **Time to keep unassigned worklist items after disappearance from DIS**, enter a time value in minutes. If a Worklist item which was retrieved from RIS in a previous query is no longer part of new DMWL answers, it is automatically deleted after the time specified in this field.

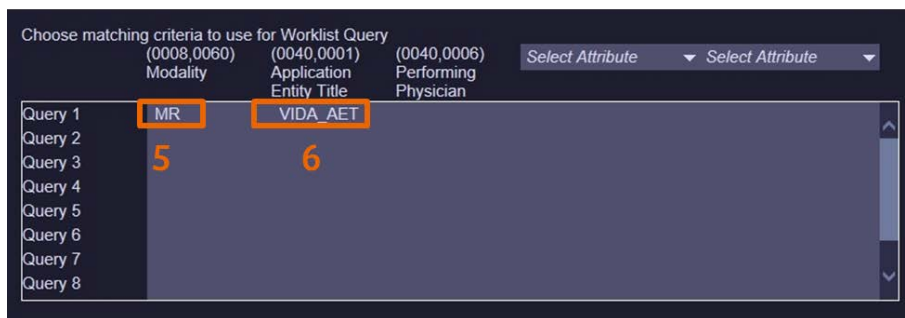
- [3] In the **Days into the past** and **Days into the future** fields, define the time interval for the **Worklist item start date**. If both values are set to **0**, the system receives only images from the current day.

To configure the **Worklist Query Attributes and their Values** [4], proceed as follows:

- Use the query table to define queries, to only retrieve Worklists for the Scanner. Typically one of the following two settings are requested:
  - Retrieve the Worklists for all MR devices
    - Enter in *Modality* "MR" [5]
  - Retrieve the Worklists that are specific to the system
    - Enter in *Modality* "MR" [5]
    - Enter in *Application Entry Title* the system specific AET [6]
- Click "Save"
- Restart the workplace. Afterwards the RIS query button in the Scheduler is available

#### Note:

Although many attributes can be selected, the RIS will only return the ones that it supports.



### 5.1.3.1 Validating Worklist

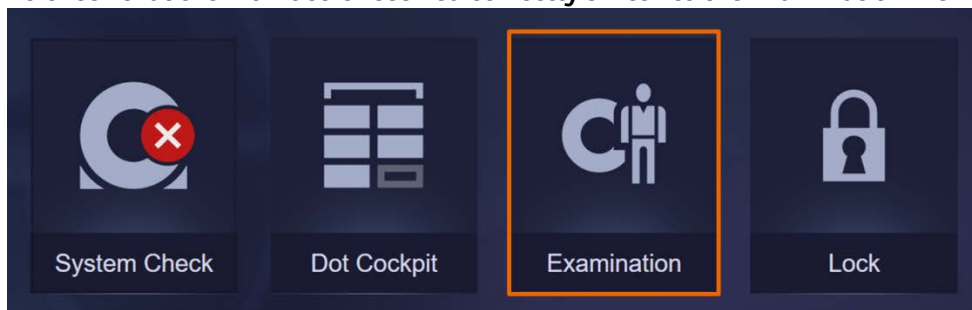
#### Hint:

To validate the Worklist settings it is recommended to temporarily set the *Time to keep unassigned worklist items after disappearance from DIS* to 1. Otherwise changes to the worklist settings that result in not showing certain patients will not be visible until the configured time setting.

To see results in the patient browser the following requirements must be fulfilled:

- The user has the rights for correct & rearrange  
(by default all users of the scanner have the necessary user rights)

To check that the Worklist is received correctly switch to the Examination View



- Click the refresh button [1] to validate the receive schedule.

The procedures according to the configured query settings should be shown

**Patient Browser**

Scheduler

Search by Name, ID, DoB, Description, Procedure Date, Acc No. and Modality (C...

**Patients** Scheduled for Today (0 result(s), 5:54:20 PM)

1 [Refresh] [View] 2

| Patient | Procedure Desc | Status |
|---------|----------------|--------|
|---------|----------------|--------|

#### Hint:

in case no Patients are displayed, check view [2] is set to *Patient View* (not *study view*)

#### Worklist Retrieval:

- Validate if the following Buttons are available after Worklist configuration



- Search for Patients using Patient Query
  - [1] set correct Source
  - [2] choose modality
  - In case Worklists are specific to the system [3]set Modality AET to local

**Patient Query**

Query from RIS node

Source: JDRIS 1

Patient ID: [Field]

Patient Name: Last Name, then First Name, then Middle Name

Modality: MR 2

Modality AET: Any 3

Start Date: 7/27/2017 M/d/yyyy

End Date: 7/29/2017 M/d/yyyy

Accession No: [Field]

Procedure ID: [Field]

Referring Physician: [Field]

Performing Physician: [Field]

Ward: [Field]

| Patient Name | Patient ID | Date of Birth | RP Description | Modality | Accession No |
|--------------|------------|---------------|----------------|----------|--------------|
|--------------|------------|---------------|----------------|----------|--------------|

Cancel

## 5.1.4 Print Settings

Local DICOM Node

Look Up Tables (LUT)

| Image Type          | Correction LUT               | Density |
|---------------------|------------------------------|---------|
| CR Image            | CR_Image_Standard            | ▼ 300   |
| CT Image            | CT_Image_Standard            | ▼ 280   |
| DX Image            | Standard_Linear              | ▼ 300   |
| DX MG Image         |                              | ▼ 300   |
| US MultiFrame Image | US_MultiFrame_Image_Standard | ▼ 240   |
| MR Image            | MR_Image_Std1Corr_16bits     | ▼ 300   |
| US Image            | US_Image_Standard            | ▼ 240   |
| SC Image            | SC_Image_Standard            | ▼ 300   |
| SA Overlay          | SA_Overlay_Standard          | ▼ 300   |
| SA Curve            | SA_Curve_Standard            | ▼ 300   |
| XA Image            | XA_Image_Standard            | ▼ 300   |
| XRF Image           | XRF_Image_Standard           | ▼ 300   |
| XA BiPlane Image    | XA_BiPlane_Image_Standard    | ▼ 300   |
| NM Image            | NM_Image_Standard            | ▼ 300   |
| PET Image           |                              | ▼ 300   |

Save Cancel

Make adjustments in alignment with hard copy print technical support engineer. In most cases it is necessary to also configure the LUT on the Printer/Print-server to optimize the contrast.

For details regarding released printer and required settings please refer to document Hardcopy Cameras which belongs to the syngo MR XA version.

## 5.1.5 General Settings

General Settings

**SCU/SCP Settings**

**Transfer Format Optimization**

☐ Uncompressed Format (low CPU load)

☒ Compressed Format (low network load)

**Connection Parameters**

Association Negotiation Timeout: 30 sec [1 - 120]

Transfer Inactivity Timeout: 30 sec [1 - 300]

TCP/IP Socket Timeout: 5 sec [1 - 60]

Maximum PDU Size: 32 KB [16 - 512]

**SCP Settings**

**Preferred AETs**

☐ Accept Only Known AE Titles

☒ Accept All AE Titles

**SCU Settings**

**Default Specific Character Set**

Select Local Language: English (ISO IR100)

☐ Use Unicode (UTF-8 / ISO IR192) for DICOM character encoding

Select this only if it is supported by all systems in your local DICOM Network

Ok Cancel

- Set **Default Specific Character Set** according to the character settings of the system and the supported characters set of the RIS/PACS system.

The following Character sets are available:

English (ISO IR6)  
English (ISO IR100)  
German (ISO IR100)  
French (ISO IR100)  
Spanish (ISO IR100)  
Japanese Katakana/Romaji (ISO IR13/14)  
Japanese Kanji (ISO IR87/159)  
Chinese (GB18030)  
Turkish (ISO IR148)  
Cyrillic (ISO IR144)  
Arabic (ISO IR127)  
Greek (ISO IR126)  
Hebrew (ISO IR138)  
Thai (ISO IR166)  
Korean KS X 1001 (ISO IR149)

### Warning:

- This setting is independent of the language settings used in the system.
- This setting has to be aligned with the clinical IT environment of the customer site (e.g. RIS/PACS). Use the Character Set which is supported/used by these systems.
- All images created by the system will use the selected Character Set.
- If the setting isn't aligned with RIS/PACS, some characters might not be displayed correctly at the MR System or at the other systems

## 5.2 Remote DICOM Node

Remote DICOM nodes are modalities, DICOM printers, RIS systems (DMWL) or other DICOM devices that can interact with the system. A set of DICOM templates supports the configuration of the remote DICOM nodes. Each template is identified by the device role, manufacturer, and model.

For a successful IT integration the DICOM Nodes details are needed of the nodes that are directly connected to the MR system via the hospital network, as well as the information on the complete hospital workflow, including any Node indirectly connected/used. For a successful IT integration it is necessary to test the settings in the complete Hospital workflow.

The following describes how to configure the Remote DICOM Nodes. Whilst not all of the configuration steps are necessary for all nodes, every node needs to be configured individually (many steps have to be repeated for each remote node).

### Warning:

When deleting a Remote node, first ensure all Autorouting , Worklist settings, which use this node, are deleted before deleting the Remote node. Failure to do so may cause problems with these configurations (e.g. Autorouting rule will not be able to be removed from the list).

### 5.2.1 Prerequisites

- **For all Remote DICOM nodes:**
  - o Manufacturer, Model, Version
  - o Logical Name, Host Name, IP Address, Port Nr.
- **Is the Remote DICOM node able to use Enhanced DICOM? (Y / N)**
  - o Identify all Systems in the customer workflow incl. Tier 2 and 3 Systems/Working Stations.

#### Hint:

eMR should only be enabled if the Remote node directly connected to the MR System and all the systems connected to this remote node (used in the customer MR workflow) are capable of eMR.

- o To check if the Remote Node is able to use Enhanced DICOM, refer to the DICOM Conformance statement of the individual Remote Node and further Systems used in the customer workflow.
  - The DICOM conformity statement can be found online (IntERnet)

#### Hint:

If the conformance statement doesn't claim conformance for a type of multi-frame object the conversion has to be enabled. But even when the conformance is claimed a dedicated test with our test images has to be performed to ensure correct display of the images at the reading workplace!

- **Does the customer use Storage Commitment (Y / N)**
  - o If customer wants to use storage commitment; check specifics of when the commitment is returned by the customers interfaces (PACS / Working Station)
- **Does the customer use MPPS? (Y/N)**



## 5.2.2 Overview

When clicking on **Remote DICOM Nodes** an overview of all configured Remote DICOM nodes is displayed. By clicking on any node name, a high level overview of the configuration is displayed.

| Diagnose                     | Repair                 | Maintenance | Installation | Technical Configuration | Quality Assurance | Clinical Configuration |
|------------------------------|------------------------|-------------|--------------|-------------------------|-------------------|------------------------|
| Site Information             | DICOM Remote Node List |             |              |                         |                   |                        |
| Remote Service               |                        |             |              |                         |                   |                        |
| Licensing                    |                        |             |              |                         |                   |                        |
| User and Role Administration |                        |             |              |                         |                   |                        |
| DICOM Nodes                  |                        |             |              |                         |                   |                        |
| Local DICOM Node             |                        |             |              |                         |                   |                        |
| Remote DICOM Nodes           |                        |             |              |                         |                   |                        |
| DICOM Data Handling          |                        |             |              |                         |                   |                        |
| MR Configuration             |                        |             |              |                         |                   |                        |
| Maintenance Configuration    |                        |             |              |                         |                   |                        |

| Logical Name       | Host Name     | IP Address     | Location | Model     |       |
|--------------------|---------------|----------------|----------|-----------|-------|
| <b>PACS</b>        |               |                |          |           |       |
| BENCTWMOD1         | BENCTWMOD1    | 192.168.51.200 |          | IntelPACS | ✓   i |
| <b>Printer</b>     |               |                |          |           |       |
| PRINTER            | Aycanprinter  | 10.1.1.1       |          | Other     | ✓   i |
| <b>RIS</b>         |               |                |          |           |       |
| JDICOMRIS          | JDICOMRIS     | 10.10.100.10   | MR       | Other     | ✓   i |
| <b>Workstation</b> |               |                |          |           |       |
| GEWORKSTA...       | SEWORKSTATION |                |          | Other     | ✓   i |

AddNew

To add a new Remote DICOM node press “AddNew” at the bottom.



### 5.2.3 Add New Remote DICOM Node

Site Information

Remote Service

Licensing

User and Role Administration

DICOM Nodes

Local DICOM Node

Remote DICOM Nodes

DICOM Data Handling

MR Configuration

Maintenance Configuration

## Add New Remote DICOM Node

Role: Please select a role

- ☐ Modality
- ☐ PACS
- ☐ Printer
- ☐ RIS
- ☐ Workstation
- ☐ teamplay
- ☐ Customized

Manufacturer:

Model:

Product Info:

Next Cancel

1. Choose role of New DICOM node
2. Choose Manufacturer
3. Choose Model
4. Press Next

#### Note:

- Your selection is used to load the appropriate configuration template. If a correct a template which fits to the characteristics of the DICOM node is not available (e.g. Print Robot), choose "Customized" from the Role list.
- If the Remote DICOM Node Manufacturer or Model is not included in the Drop down, choose "Other" from Drop Down. Subsequently inform the RSC of missing Manufacturer / Model.
- The list of supported Hard copy devices can be found in the Software Information for Service - syngo MR XA... or the TDOC Hardcopy Cameras which belongs to the syngo MR XA version. Refer to Chapter 3.3 to configure Burn Robots.
- teamplay: a teamplay remote DICOM node needs to be configured when using a teamplay server images or teamplay analytics. Details on the configuration can be found in Chapter 3.4.

## 5.2.4 Remote Node Configuration Overview

Depending on which Remote DICOM node Role is selected and depending on the Manufacturer and Model different services will be available / shown from the start (e.g. [10], [11] etc.).

To adjust the AET and Port information for a specific Service the unlock button is pressed [14]

To remove a Service: all AET information **and** the Port information need to be deleted before saving. The following Services are available:

### **Storage Service:**

The DICOM Storage Service transfer images and other objects from and to remote systems

### **Query/Retrieve:**

The DICOM Query/Retrieve Service provides the possibility to request a list of patient data (query) from a remote system. Based on the provided list one can select patients/studies to be transferred to the local system (retrieve).

### **Storage Commitment:**

The DICOM Storage Commitment Service sends a list of all transferred images to the destination system. The response from the remote systems commits that the images are successfully stored/archived. Usually Storage Commitment is used for archive destinations.

### **MPPS:**

Modality Performed Procedure Step (MPPS) Service sends the information how the procedure was performed back to the remote node (e.g. contrast agent, time/status tracking, list of series, protocol and images)

### **Worklist:**

The Worklist Service automatically transfers the master index of patient demographics and exam scheduling information from any HIS/RIS.

### **Print:**

The Print Service is used to send DICOM images directly to a hard copy print device.

### **UpsWatch**

Unified Procedure Step that queries for worklist items of interest, gets details for a specific worklist item, submits cancellation request for a specific worklist item, subscribes for an event notification of all worklist items or an event notification for changes to a certain worklist item

### **UpsPush**

Unified Procedure Step that creates a new worklist item on a worklist or submits a cancellation request for a worklist item



To configure the Remote DICOM node additional information (see prerequisites 5.2.1) needs to be received from the local Network admin, the MTA and the Radiologist.

1. Enter Remote Node information in
  - [1] Enter the host name of the DICOM node.
  - [2] Enter the IP address of the DICOM node
  - [3] Enter a name which will be used to identify the DICOM node in the user interface
  - In the "Location" field, enter the physical location of the DICOM node (optional).
2. In case AET (application entity titles) does not differ for Storage, Query Retrieve, etc. press Default AET. By doing so the Logical Name will be copied to all Ports.
3. Enter/edit Port information [6]
4. SAVE!
5. Perform Validation as described in 5.2.4.1
6. Depending on the Type of DICOM node the following services need to be configured if available/requested by customer by clicking  at the corresponding Service
  - Adjust Storage Settings [10] (5.2.5)
  - Adjust Storage Commitment Settings [12] (5.2.7)
  - Adjust MPPS (5.2.8)
  - Adjust Print [13] (5.2.9)
7. Validate each configured Service as described in the following chapters

**Hint:**

- The changes take effect immediately. A system restart is not required.
- When the host name is known at the WINS or Domain Name Server (DNS). Click the **nslookup** [8]. If the name is found, the corresponding IP address is automatically filled in with the resolved value.
- Either host name or IP address must be provided. If both are provided, the IP address has priority.

### 5.2.4.1 Validating IP Address and Port / AET

To validate if communication is possible between the System and the Remote DICOM Node

- Test(ping): Checks if the IP Address communication is possible
- Test(C-Echo): Checks if the Port and AET is communicating

**Hint:**

even if the IP ping fail, verify if C-Echo is working

### 5.2.4.2 Troubleshooting IP Address and Port / AET

If errors occur on these tests the following root causes are possible:

- Wrong IP Address
- Wrong AET
- Wrong Port
- On the Remote DICOM Node our system is not configured / communication from our system isn't enabled

## 5.2.5 Storage Settings

DICOM Remote Node Editor > Storage Settings

Storage Send Properties (SCU) Storage Receive Properties (S...

Properties for sending to node: TESTWORKSTATION (no archive)

Transfer Syntaxes to be Excluded

| SOP Class Name                                       | SOP Class UID               | Unsaved Changes |
|------------------------------------------------------|-----------------------------|-----------------|
| ▶ All SOP Classes                                    | *                           |                 |
| ▶ Computed Radiography Image                         | 1.2.840.10008.5.1.4.1.1.1   |                 |
| ▶ Digital X-Ray Image - for Presentation             | 1.2.840.10008.5.1.4.1.1.1.1 |                 |
| ▶ Digital X-Ray Image - for Processing               | 1.2.840.10008.5.1.4.1.1.1.1 |                 |
| ▶ Digital Mammography X-Ray Image - for Presentation | 1.2.840.10008.5.1.4.1.1.1.2 |                 |
| ▶ Digital Mammography X-Ray Image - for Processing   | 1.2.840.10008.5.1.4.1.1.1.2 |                 |
| ▶ Digital Intra-Oral X-Ray Image - for Presentation  | 1.2.840.10008.5.1.4.1.1.1.3 |                 |

Header Correction Ruleset

! These changes may affect system interoperability and DICOM conformance. We strongly recommend testing communication with the selected node for correctness before setting it to productive.  
If encapsulation is enabled, header correction will be disabled for non-supported objects.

Applied Header Correction Rules

Available Header Correction Rules

Handle Acquisition Time  
Handle Patient's Age  
Handle Patient's Sex  
Handle Patient's Weight and Size  
Normalize Patient's Age  
Remove All Private Data Elements

Storage Commitment

1

☒ Disable

! If you use a node without storage commitment ("No SC") as archive node, you will not receive IHE-compliant confirmation of archiving. The archive state is only updated based on the transfer result. Permanent data storage is not ensured and data loss may result. Check whether changes affect autorouting rules or the archive configuration.

Destination: Same as Storage

Encapsulate Non-Supported Objects

2

☒ Enable

i

If you enable this option, non-supported objects will be encapsulated in the "Basic Text Structured Report" or "Secondary Capture Image" format depending on the user's selection. Further processing of sent images is possible after images are retrieved to the syngo.via system. Check whether changes affect autorouting rules or header correction rules.

Select Format:

☐ Basic Text Structured Report  
☐ Secondary Capture Image

MR Export Rules

MR Image Conversion

3

☒ Convert GSPS for MR Images

Separate Series for MR and SC Color

Multiframe Conversion

☒ Convert Enhanced MR and related GSPS  
☒ Exclude BOLD Images

Separate Series for MR and SC Color

☒ Convert Enhanced MR Color LUT and related GSPS

Separate Series for MR and SC Color

☒ Convert Enhanced MR Color

☒ Convert Multiframe Color SC

☒ Convert Multiframe Grayscale Word SC and related GSPS

i

Do not convert Enhanced MR Images for remote syngo.via nodes because of interoperability reasons

Multiframe Conversion Result Structure

4

☒ Preserve original Series Number for each converted Multiframe Image

☐ Separate Series for each converted Diffusion Multiframe Image

☐ Separate Series for each converted Multiframe Image (all other than Diffusion, e.g. Twist, Grasp, ...)

5

OK

Cancel



Usually the "Transfer Syntaxes to be excluded" and "Header Correction Ruleset" doesn't need to be adjusted. Only if deeper DICOM communication problems occur adjustments may be necessary. If this is the case contact the RSC.

### 1. **Storage Commitment [1]**

**Recommended setting: "Disabled" (Checked)**

- Depending on the Remote DICOM Node and the customer preference enable/disable Storage Commitment.
- In case Storage Commitment is Enabled the Service also needs to be configured in the Remote DICOM Node (see 5.2.4 and 5.2.7) and the correct Destination needs to be selected from the Drop down. **Recommended setting: "Same as Storage"**

### 2. **Encapsulate Non-Supported Objects [2]**

**Recommended Setting: "Not Enabled" (not checked)**

#### **Hint:**

When enabled, encapsulation will always be active for the complete list. If any of these are needed in the native format in the PACS, encapsulation should not be used. Most of the objects are created only from syngo.via Advanced Applications (→ AV objects).

When enabled the following Objects will be encapsulated:

- Registration SOP class (1.2.840.10e08.5.1.4.1.1.66.1) → **AV object**
- Segmentation SOP class (1.2.840.10008.5.1.4.1.1.66.4) → **AV object**
- Surface Segmentation (1.2.840.10008.5.1.4.1.1.66.5) → **AV object**
- Encapsulated PDF SOP class (1.2.840.10008.5.1.4.1.1.104.1) → **syngo.via Report**
- Structured Report SOP class basic (1.2.840.10008.5.1.4.1.1.88.11) → **Evidence document**
- Structured Report enhanced (1.2.840.10008.5.1.4.1.1.88.22) → **PhoenixZIP**
- Structured Report SOP class comprehensive (1.2.840.10008.5.1.4.1.1.88.33) → **Evidence document**
- Real World Value Mapping SOP class (1.2.840.10008.5.1.4.1.1.67) → **AV object**
- MR Spectroscopy SOP class (1.2.840.10008.5.1.4.1.1.4.2) → **Spectroscopy data**
- DICOM Raw Data SOP class (1.2.840.10008.5.1.4.1.1.66) → **DTI Tensor**

**Encapsulate Non-Supported Objects**

☒ **Enable**

 If you enable this option, non-supported objects will be encapsulated in the "Basic Text Structured Report" or "Secondary Capture Image" format depending the user's selection. Further processing of sent images is possible after images are retrieved to the syngo.via system. Check whether changes affect autorouting rules or header correction rules.

**Select Format:**

☒ **Basic Text Structured Report**

☐ **Secondary Capture Image**

### 3. MR Export Rules [3]

Case the customer is **not** using enhanced DICOM:

- Check all Boxes in [1] MR Image Conversion and Multiframe Conversion → **Interoperability Mode**
- **Recommendation:** Select “Separate Series for MR and SC Color” from Dropdown[2]
- The “Multiframe Conversion Result Structure” should be clarified with the customer (PACS, reading requirements)  
**Recommendation:**
  - Select “Preserve original Series Number for each converted Multiframe Image”
  - Don't select “Separate Series for each converted Diffusion Multiframe Image”
  - Don't select “Separate Series for each converted Multiframe Image (all other than Diffusion, ...)”
- Details on the effects of the export rules can be found in the next chapter
- Go to Step 4

The screenshot shows the 'MR Export Rules' dialog box. It is divided into several sections. The first section, 'MR Image Conversion', contains a checked checkbox 'Convert GSPS for MR Images'. The second section, 'Multiframe Conversion', contains several checked checkboxes: 'Convert Enhanced MR and related GSPS', 'Exclude BOLD Images', 'Convert Enhanced MR Color LUT and related GSPS', 'Convert Enhanced MR Color', 'Convert Multiframe Color SC', and 'Convert Multiframe Grayscale Word SC and related GSPS'. To the right of these checkboxes is a dropdown menu with the text 'Separate Series for MR and SC Color'. The third section, 'Multiframe Conversion Result Structure', contains three checkboxes: 'Preserve original Series Number for each converted Multiframe Image' (checked), 'Separate Series for each converted Diffusion Multiframe Image' (unchecked), and 'Separate Series for each converted Multiframe Image (all other than Diffusion, e.g. Twist, Grasp, ...)' (unchecked). At the bottom of the dialog are 'OK' and 'Cancel' buttons. Three orange boxes with numbers 1, 2, and 3 highlight specific areas: box 1 highlights the 'Multiframe Conversion' section, box 2 highlights the dropdown menu, and box 3 highlights the 'Multiframe Conversion Result Structure' section.

**MR Export Rules**

**MR Image Conversion**

- ☒ Convert GSPS for MR Images

**Multiframe Conversion**

- ☒ Convert Enhanced MR and related GSPS
- ☒ Exclude BOLD Images
- ☒ Convert Enhanced MR Color LUT and related GSPS
- ☒ Convert Enhanced MR Color
- ☒ Convert Multiframe Color SC
- ☒ Convert Multiframe Grayscale Word SC and related GSPS

Do not convert Enhanced MR Images for remote syngo.via nodes because of interoperability reasons

**Multiframe Conversion Result Structure**

- ☒ Preserve original Series Number for each converted Multiframe Image
- ☐ Separate Series for each converted Diffusion Multiframe Image
- ☐ Separate Series for each converted Multiframe Image (all other than Diffusion, e.g. Twist, Grasp, ...)

OK Cancel

Case the customer is **using enhanced DICOM**:

- Uncheck all boxes → **enhanced Mode**
- Details on the effects of the export rules can be found in the next chapter
- Go to Step 4

The screenshot shows the 'MR Export Rules' dialog box. It has a dark blue background with white text. The dialog is divided into several sections. The first section, 'MR Image Conversion', contains a checkbox 'Convert GSPS for MR Images'. The second section, 'Multiframe Conversion', contains several checkboxes: 'Convert Enhanced MR and related GSPS' (with a sub-option 'Exclude BOLD Images'), 'Convert Enhanced MR Color LUT and related GSPS', 'Convert Enhanced MR Color', 'Convert Multiframe Color SC', and 'Convert Multiframe Grayscale Word SC and related GSPS'. A large orange box labeled '1' highlights this entire section. To the right of this section are three dropdown menus, each with the text 'Separate Series for MR and SC Color' and a downward arrow. A large orange box labeled '2' highlights these three dropdown menus. Below the 'Multiframe Conversion' section is an information icon (i) and a text message: 'Do not convert Enhanced MR Images for remote syngo.via nodes because of interoperability reasons'. The third section, 'Multiframe Conversion Result Structure', contains three checkboxes: 'Preserve original Series Number for each converted Multiframe Image', 'Separate Series for each converted Diffusion Multiframe Image', and 'Separate Series for each converted Multiframe Image (all other than Diffusion, e.g. Twist, Grasp, ...)'. A large orange box labeled '3' highlights this section. At the bottom of the dialog are 'OK' and 'Cancel' buttons.

#### 4. **Validate Storage Settings**

- see 5.2.5.2

#### **Hint:**

- To enable Storage commitment both the check mark here needs to be unchecked and the service in the Remote Node needs to be configured



### 5.2.5.1 MR Export Rules

**Note:**

- The DTI Tensor data itself is stored as Raw Data (1.2.840.10008.5.1.4.1.1.66).
- The Spectroscopy data is stored as MR Spectroscopy (1.2.840.10008.5.1.4.1.1.4.2)
- For these two there is NO conversion via the DICOM Export Rules available, as there are no “old” DICOM formats available as conversion target.  
In the past these were stored completely private as “CsaNonImages”, which was rarely was accepted by PACS.

Refer to Chapter 2 for reference of Abbreviations used in the following.

#### **MR Image Conversion**

- **Convert GSPS for MR images**  
MRImage+PR → MRImage (with overlay Bitmap)  
Example:  
PR Graphics will be converted to Bitmap (burned in)

#### **Multiframe Conversion**

- **Convert Enhanced MR and related GSPS**  
eMR + PR → MRImage incl. Overlay (b/w graphics)  
MRImage + SC (color graphics)  
Example:  
“normal” acquisition output, Inline VF, Segmentations
  - **Exclude BOLD Images**  
BOLD Images excluded from Rule above.  
**Hint:**  
Due to the large amount of images created when converting Enhanced MR and related GSPS for Bold images, it is recommended to exclude Bold images when activating the this conversion Rule.
- **Convert Enhanced MR Color LUT and related GSPS**  
eMR + PR → MRImage + SC  
Example:  
Perfusion Maps, T2 MAPs
- **Convert Enhanced MR Color**  
eMR → SC  
Example:  
Diffusion Color FA Maps  
  
Enhanced MR Color Images (1.2.840.10008.5.1.4.1.1.4.3) can be converted to Secondary Capture Images (1.2.840.10008.5.1.4.1.1.7). In that case with RGB Pixel representation. These images are the FA Color Maps from DTI.
- **Convert Multiframe Color SC**  
MF C SC → SC  
Example:  
*Currently not used*

- **Convert Multiframe Greyscale Word SC and related GSPS**  
MF SC → SC  
Example:  
Evaluation, Graphs, Tables,....
- **Dropdown options:**
  - **All Images in one Series**  
MR Image + SC Snapshot are in one image
  - **Separate Series for MR and SC Color**  
MR Image and SC Snapshot are in separate Series
  - **Store SC Color Images Only**  
Only available for “Convert Enhanced MR Color LUT and related GSPS”  
MR Images are not stored; only SC Snapshot are stored

### **Multiframe Conversion Result Structure**

The options in this frame define the image Series numbering.

- **Preserve original Series Number for each converted Multiframe Image**  
The Series numbering equals that of the Original eMR Image. In case of the Series containing several Multiframes, the series number is the same for each converted Multiframe Image  
  
**Note:** In most cases this option is the desired behavior, but might lead to duplicate Series Numbers if “Separate Series for each converted...” is activated. Clarify with Radiologists reading the image.
- **Separate Series for each converted Diffusion Multiframe Image**  
In case of multiple Diffusion B-values are stored within one Series this option defines if the Series is split into separate Series per Multiframe instance.  
Scope: B-Values  
Consideration:  
Capabilities: Browser, Viewer (navigation through B-values, loading to one/separate segments)
- **Separate Series for each converted Multiframe Image (all other than Diffusion, e.g. Twist, Grasp, ...)**  
In case of more than one eMR instance per series this option defines if the Series is split into separate Series per Multiframe instance. Not affecting Diffusion Images  
Examples: Echo Times, Timepoints, ..  
Consideration:  
Capabilities: Browser, Viewer (navigation through timepoints)  
Pro: identification of a certain timepoint  
Con: large amount of Series in case of many timepoints

### Example of the effects on image Series numbering:

| Dropdown options                    | Original eMR Series Number               | Separate Series for each converted Multiframe                       |        |                                        |                  |
|-------------------------------------|------------------------------------------|---------------------------------------------------------------------|--------|----------------------------------------|------------------|
|                                     |                                          | No                                                                  |        | Yes                                    |                  |
|                                     |                                          | Preserve original Series Number for each converted Multiframe image |        |                                        |                  |
| No impact                           |                                          | No                                                                  | Yes    | No                                     | Yes              |
|                                     | 1                                        | 1001                                                                | 1      | 1001                                   | 1                |
|                                     | 2                                        | 2001                                                                | 2      | 2001                                   | 2                |
|                                     | 3                                        | 3001                                                                | 3      | 3001                                   | 3                |
|                                     | 4 (4 Multi-frames)                       | 4001                                                                | 4      | 4001<br>4002<br>4003<br>4004           | 4<br>4<br>4<br>4 |
|                                     | 5 (2 Multi-frames)                       | 5001                                                                | 5      | 5001<br>5002                           | 5<br>5           |
| Separate Series for MR and SC Color | 6 (1 Multi-frame with Color PR graphics) | 6001 (MR Images)<br>6001 (SC Snapshot)                              | 6<br>6 | 6001 (MR Images)<br>6001 (SC Snapshot) | 6<br>6           |
| All Images in one Series            |                                          | 6001 (MR Images + SC Snapshot)                                      | 6      | 6001 (MR Images + SC Snapshot)         | 6                |

## 5.2.5.2 Validating Storage Settings

- Validate Basic Storage
  - Store image to Remote node
  - Check Remote node for correct transfer
- Validate Job Queue
- Validate Browser Flags
  - Check: Exported flag set for non-archive nodes.
  - Check: Archived flag set for archive nodes.

**Note:**  
if storage commitment is used state transition from “Queued for archiving” to “Archived” is only done after commitment had been received.
- Interoperability Test:
  - Send test image to Remote Node (instructions can be found in Chapter 0)
  - Validate correct display of DICOM test images

## 5.2.5.3 Troubleshooting Storage Settings

- In case of communication problems with the remote node
  - Contact RSC
  - DICOM trace may be necessary – refer to Chapter 8.1.1 for details.
- In case of interoperability problems: please contact RSC

## 5.2.6 Query Retrieve

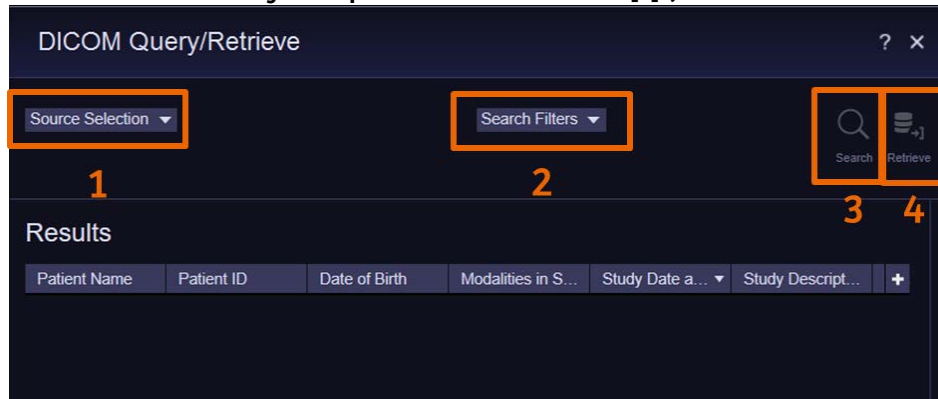
No configuration is necessary for Query / Retrieve, but it is important to validate its functionality.

### 5.2.6.1 Validating Query / Retrieve

1. in case no DICOM data available from PACS, send DICOM Test Data to Remote DICOM Node
2. Navigate to **Patient Browser** → **DICOM Q/R**



3. select PACS system from **Source Selection** [1] (otherwise Search button will not be available)



4. Set Search Filters [2] (part of the Patient Name and the Modality needs to be entered otherwise no data will be found!, e.g. MR\* for Patient name)

A screenshot of the 'Search Filters' dialog box. It contains several input fields:   
- 'Patient Name' with a placeholder text: 'Last Name, then First Name, then finally Middle'.   
- 'Patient ID' with an empty text field.   
- 'Accession Number' with an empty text field.   
- 'Study Date and Time' with a '<Custom>' dropdown, a 'From' date field, a 'To' date field, and a checked 'All day' checkbox.   
- 'Study Description' with an empty text field.   
- 'Modalities in Study' with a dropdown menu.   
At the bottom, there are three buttons: 'Search' (magnifying glass), 'Remove All' (X), and 'Add Condition' (+).

5. press **Search**
6. Select file and press Retrieve
7. Confirm successful transfer of DICOM images in Patient Browser

### 5.2.6.2 Troubleshooting Query / Retrieve

- Ensure that the scanner is correctly configured at the remote node to use Query / Retrieve as a SCU
- Ensure that the remote node is correctly configured at the Scanner (AET, Port, etc.)
- Remove DICOM Node, add new DICOM Node, remove Query/Retrieve from node configuration (it may be necessary to press **unlock** to remove Port ID and AET)

### 5.2.7 Storage Commitment Settings

**Hint:**

- If PACS is set up to send "Storage commitment" response only during a defined time period (e.g. overnight) and the scanner does not get the this commitment in a defined time period, it may cause the Archiving state to show up as "failed" even though the data has been successfully transferred.
- This service does not have to be removed to deactivate Storage commitment. Disabling it in the Storage settings is sufficient (Chapter 5.2.5)

DICOM Remote Node Editor > Storage Commitment Settings

Storage Commitment Send Properties (SCU)

**Timeouts and Retries**

|                   |    |                 |
|-------------------|----|-----------------|
| Delay Time        | 10 | min [0 - 14400] |
| Result Timeout    | 60 | min [1 - 1440]  |
| Number of Retries | 3  | [1 - 100]       |

**Sending of N-Action**

This setting determines whether the system will send an N-Action for each object separately or bundled per transfer job.

☒ Bundled Call

OK Cancel

**1. Adjust the following settings according to Remote Node configuration**

- Delay Time
- Result Timeout
- Number of Retries (**Recommended setting: 3**)

**Example:**

If the PACS only sends the Storage commitment response every 4 hours, adjust the delay time accordingly

## 5.2.7.1 Validating Storage Commitment

1. Navigate to the **Patient Browser**
2. Right click on DICOM Test Image and select **send to Archive**
3. Check Status of Study Archived
  - Archiving State turns from “Queued for archiving” to “failed” [1]
  - Archiving State turns from “Queued for archiving” to “archived” [2]
  - Archiving State “to be Archived” [3]
  - Archiving state “not to be Archived” [4]

### Hint:

The Status change can last until the configured Timeout

Patient Browser

Local Data +

Search by Name, ID, Date, Description and Modality (Ctrl+F)  More Filters

Results (14 results, 4:50:34 PM)

| Patient Name   | Patient ID      | Date of Birth | Study Date a...  | Study Descript...  | Accession Nu... | Workflow | Study Archived                                                                        | Corr + |
|----------------|-----------------|---------------|------------------|--------------------|-----------------|----------|---------------------------------------------------------------------------------------|--------|
| SiemensTest... | 123456-SMR...   | 7/2/1973      | 5/18/2017 3...   | whole body o...    |                 |          |    |        |
| SiemensTest... | 3333-SMR-7...   | 4/24/1971     | 5/18/2017 1...   | breast Breast...   |                 |          |    |        |
| SiemensTest... | 2017.05.18-1... | 1/1/1947      | 5/18/2017 10...  | Head clinical...   |                 |          |    |        |
| SiemensTest... | 2017.05.18-0... | 1/1/1967      | 5/18/2017 9...   | Abdomen Ab...      |                 |          |   |        |
| SiemensTest... | 2017.03.06-1... | 3/6/1973      | 3/6/2017 12...   | head clinical I... |                 |          |  |        |
| SiemensTest... | SP              | 6/8/1957      | 3/3/2017 11:0... | heart Cardiac...   |                 |          |  |        |
| MR_Onco_Li...  | E11_C39         | 12/1/1964     | 8/5/2013 2.0...  | Liver_201305...    |                 |          |  |        |
| MR_Onco_Li...  | Skyra_Liver...  | 1/25/1982     | 6/20/2013 12...  | Liver_201306...    |                 |          |  |        |
| MR_L-Spine...  | Skyra_LSpin...  | 5/1/1948      | 6/20/2013 10...  | LSpineDot_2...     |                 |          |  |        |
| MR_C_Spine...  | 123             | 1/1/1985      | 9/21/2012 2...   | Demo_Spine...      |                 |          |  |        |
| MR_Onco_Li...  | Siemens         | 3/2/1947      | 6/14/2012 5...   | 1st_Follow_up      |                 |          |  |        |
| MR_Tissue4...  | Skyra_Prosta... | 5/1/1947      | 10/13/2011 1...  | Prostate_201...    |                 |          |  |        |
| MR_Cardiac...  | Skyra           | 6/22/1952     | 7/14/2011 10...  | Infarct            | HERZ AMB        |          |  |        |
| MR_W-Spine...  | WholeSpine      | 8/25/1960     | 8/23/2010 3...   | WholeSpine...      |                 |          |  |        |

### Note:

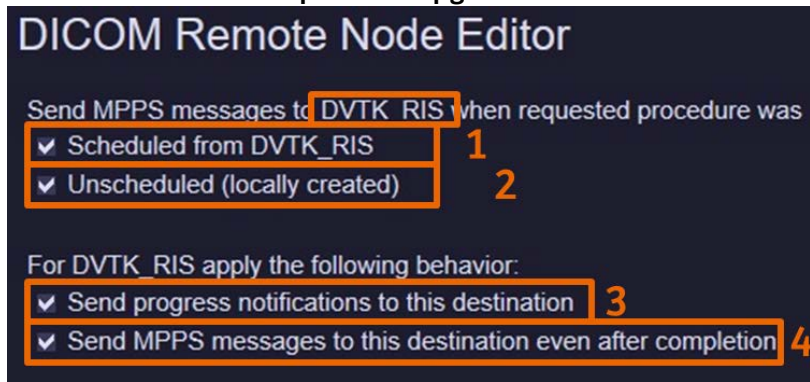
If a study is marked auto-archived, exported to an archive or autorouted to an archive, it gets flagged with the same “Archived” state.

### 5.2.7.2 Troubleshooting Storage Commitment

- Check Storage commitment is configured on Remote Archiving Node
- Check commitment confirmation settings on Remote Archiving Node is in line with the settings in Storage Commitment (especially delay time settings)

### 5.2.8 MPPS Settings

MPPS services are used for RIS configurations



- [1] Set if MPPS messages shall be sent to the remote node  
→ general activation
- [2] Set if MPPS messages shall be sent for locally created patients  
Hint: deactivate if RIS has problems with unscheduled patients
- [3] Deactivate if progress notifications after each scan cause problems with remote node
- [4] Deactivate if append MPPS cause problems with the remote node  
→ If a MPPS is already completed, but further/additional Scans are required, then a second MPPS will be appended and communicated to RIS.

**Recommended Setting:** all options activated (checked).

#### 5.2.8.1 Validating MPPS Settings

- Perform Dummy scan
- Validated correct transfer on the remote node (RIS admin)

## 5.2.9 Print Settings

### Note:

Barten LUT is not released for MR use

### Hint:

Make adjustments in alignment with hard copy print technical support engineer; adaptations may be necessary at the printer (LUT configuration)

For details regarding released printer and required settings please refer to document Hardcopy Cameras which belongs to the syngo MR XA version.

The following settings need be configured in this dialog:

- Composing Parameter: LUT mode, printing bit depth, and viewing room lighting parameters
- Timeout & Delay: Timing parameters for status polling and inactivity timeout.

### DICOM Remote Node Editor

#### Composing Parameter

Attached LUT Mode CorrectionLut

☐ 12 bit support

☒ 8 bit support

Illumination  cd/m2

Reflected Ambient Light  cd/m2

#### Supported Films/Papers

| Available                           | Name       | Portrait<br>Width (pixel) | Portrait<br>Height (pixel) | Landscape<br>Width (pixel) | Landscape<br>Height (pixel) | Medium   |
|-------------------------------------|------------|---------------------------|----------------------------|----------------------------|-----------------------------|----------|
| <input checked="" type="checkbox"/> | Inch 14x17 | 4488                      | 5464                       | 5464                       | 4488                        | BlueFilm |
| <input checked="" type="checkbox"/> | Inch 14x14 | 4488                      | 4488                       | 4488                       | 4488                        | BlueFilm |
| <input checked="" type="checkbox"/> | Inch 11x14 | 3514                      | 4488                       | 4488                       | 3514                        | BlueFilm |
| <input checked="" type="checkbox"/> | Inch 10x12 | 3184                      | 3828                       | 3828                       | 3184                        | BlueFilm |
| <input checked="" type="checkbox"/> | Inch 8x10  | 2528                      | 3184                       | 3184                       | 2528                        | BlueFilm |

#### Supported Mediums

| Name      | Color                               | Pixel Size<br>Supported Width (µm) | Pixel Size<br>Height (µm) | Min Density | Max Density | Background<br>Color |
|-----------|-------------------------------------|------------------------------------|---------------------------|-------------|-------------|---------------------|
| BlueFilm  | <input checked="" type="checkbox"/> | 78.154                             | 78.154                    | 20          | 300         | Black               |
| ClearFilm | <input checked="" type="checkbox"/> | 78.154                             | 78.154                    | 20          | 300         | Black               |

#### Timeout & Delay

Printer Status Polling Interval  300 Sec. [60-3600]

Inactivity Timeout  60 Sec. [5-180]

OK

Cancel



### **Configuring composing parameters**

From the "LUT Mode" selection list, select "Correction LUT", "Barten LUT", or Linear LUT.

In the "Composing Parameter" area, select 12 bit support or 8 bit support. This value determines the bit depth of image data send to the printer.

In the Composing Parameter area, you can specify the lighting conditions in the viewing room. The system only sends these parameters to the printer if the "Barten LUT" option is active.

In the "Illumination" text field, enter a value for the brightness of the light box in candelas per m2. This value represents the luminance of the lightbox illuminating the film. In case of paper film, it represents the luminance obtained by diffuse reflection of the illumination present in the viewing room. A higher Illumination value will result in a darker print.

In the "Reflected Ambient Light" text field, enter a value for the brightness in the viewing room in candelas per m2. For transmissive film, this value represents the luminance obtained by reflected light. A higher ambient light value will result in a higher contrast of the print.

### **Adjusting the formats of each printer medium**

In the Supported Films/Papers area, select the parameters of the media for the specific printer. The films and papers defined here will be shown in the system for the current printer. The width and height parameters define the size of the matrix that will be generated by the composing application. In the line containing the desired name, select the available check box to make a medium selectable in the system.

In the Landscape Width (pixel), Landscape Height (pixel), Portrait Width (pixel), and Portrait Height (pixel) text fields, enter the dimensions and the orientation for each medium.

In the "Medium" list, select the type of each medium, for example BlueFilm, ClearFilm, or Paper.

#### **Note:**

It should only be necessary to remove formats not used by the customer. If formats are missing contact Siemens Service / RSC

### **Adjusting supported media**

In the Supported Mediums area, you can view the Name of each medium. The Color Supported check box shows if the medium can be color printed on the current printer. The printing application will generate a colored image when this medium is used. The Pixel Size Width (µm) and the Pixel Size Height (µm) text fields show the physical pixel size of a printer. This values define the number of pixels on a sheet. They cannot be changed.

In the "Min Density" and "Max Density" text fields, enter the minimal density and the maximum density.

### **Adjusting timeout and delay settings**

In the Timeout & Delay area, adjust the Printer Status Polling Interval and Inactivity/Timeout text fields.

In the Printer Status Polling Interval text field, enter a value in seconds.

This value represents the time the system waits between sending two status requests to the printer.

In the Inactivity/Timeout text field, enter a value in seconds.

This value represents the time without any data transfer after that the system closes the DICOM association to a printer.

## **5.2.9.1 Validating Print**

To validate the Print settings, Print a selection of DICOM test images and verify print quality.

For details refer to document Hardcopy Cameras which belongs to the syngo MR XA version.

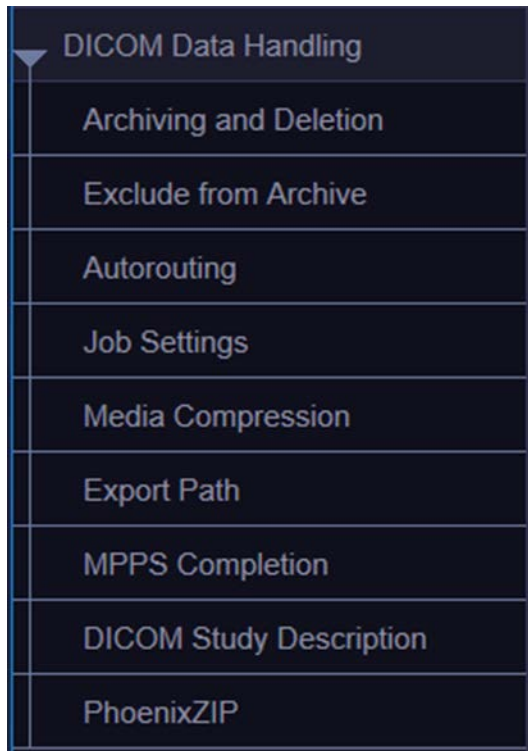
## 6 DICOM Data Handling

DICOM Data Handling can be accessed using the following path:

Administration Portal → Technical Configuration → DICOM Data Handling

Or

Administration Portal → Clinical Configuration → DICOM Data Handling



|                         |
|-------------------------|
| ▼ DICOM Data Handling   |
| Archiving and Deletion  |
| Exclude from Archive    |
| Autorouting             |
| Job Settings            |
| Media Compression       |
| Export Path             |
| MPPS Completion         |
| DICOM Study Description |
| PhoenixZIP              |

### 6.1 Prerequisites

- Does the customer want Images to be automatically archived?
  - If yes, which images, and where to?
- Does the customer want images to be preselected for archiving?
- Does the customer want Images to be AutoRouted to specific DICOM nodes? (Y/N)
  - In case of Yes
    - Where shall the images go?
    - Shall different images be sent to different nodes?
    - Shall some images be excluded from the auto routing?

### 6.2 Archiving and Deletion

The two following scenarios are the most common and we will first describe how to configure them.

Scenario 1: The customer has no PACS. Archiving should be turned off in the system.

Scenario 2: The customer has a PACS

### 6.2.1 Scenario 1 – Customer has NO PACS

The customer has no PACS. Archiving should be turned off. To turn off archiving:

1. Login to the administration portal
2. Navigate to **Technical Configuration > DICOM Data Handling> Archiving and Deletion**
3. Select **Media-based ( No PACS is configured, manual archiving is advised)**

The screenshot shows the 'Archiving' configuration page. At the top, there are two radio buttons: 'Media-based (No PACS is configured, in this case manual archiving is advised)' which is selected and highlighted with an orange border, and 'PACS-based'. Below this, there is a section titled 'Use the following DICOM Nodes as Archives' with a list box containing 'TESTWORKSTATION' and 'TESTPACS (no SC)', where 'TESTPACS (no SC)' is checked. To the right of this list box is a yellow warning icon and a text box stating: 'Nodes without Storage Commitment ("no SC") do not provide confirmation of archiving. Therefore permanent data storage is not ensured, what may result in data loss.' Below the list box is a 'Default Archive' dropdown menu. At the bottom, there is a checkbox for 'Enable automatic archiving' which is unchecked. Below this are two time selection fields: 'Start automatic archiving at' set to '01:00' and 'Stop scheduling archive jobs at' set to '02:00'.

**Warning:** The system will not send any data to a PACS. BUT if automatic deletion is enabled data will be deleted, according to the rules for automatic deletion even if the data has not been archived on film, removable disc or another media. Data will be lost if it is not saved manually.

**Note:** it is not recommended to activate auto-deletion at the scanner.

## 6.2.2 Scenario 2 – Customer has a PACS

**Archiving**

- ☐ Media-based (No PACS is configured, in this case manual archiving is advised)
- ☒ PACS-based

Use the following DICOM Nodes as Archives

- ☐ TESTWORKSTATION
- ☒ TESTPACS (no SC)

Default Archive: TESTPACS (no SC)

☒ Enable automatic archiving

Start automatic archiving at: 01:00

Stop scheduling archive jobs at: 02:00

**Automatic Deletion**

- ☐ Disable automatic deletion (No scheduled deletion is performed, in this case manual deletion is advised)
- ☒ Enable automatic deletion

Check STS fill level every: 30 minutes

Scheduled deletion above: 80.00 % (low watermark)

Start scheduled deletion at: 04:00

Immediate deletion above: 85.50 % (high watermark)

Start deletion with:
 

- ☒ least used images
- ☐ oldest stored images

**Short Term Storage (STS)**

High watermark 85.50%  
Low watermark 80.00%  
DICOM Stop

Total STS Size: 442.25 GB  
Free Space: 373.3 GB  
Current fill level: 15.59 %

Save

1. Login to the administration portal
2. Navigate to **Technical Configuration > DICOM Data Handling> Archiving and Deletion**
3. Select **PACS-based**
4. Select all DICOM nodes that will be involved in archiving [2]
5. Depending on the customer preference select a default archive from the drop-down box. [3]

### Hint:

- Setting a **"Default Archive"** preselects all images in the **"Distribution Step"** of **"MR View&GO"** and this PACS appears at the most left position in the distribution list. Images that shall not be archived need to be manually unchecked.
- Setting a **"Default Archive"** enables appearance of **„Send to Archive"** button in browser.

## 6. Automatic Archiving settings

- If customer wants images to be automatically archived during a defined time period this needs to be enabled ["4"] (**not recommended**)
- In case Auto Archiving is enabled, "Exclude from Archive" needs to be configured (see chapter 6.3)

**Important:** Auto-archiving will only work if the desired data sets are manually set to "Mark for Archiving" in the Patient Browser. By default all data sets are set to "Prevent from archiving" and thus will be excluded from archiving. Auto-archiving and its exclusions add high complexity to the set-up

## 7. Automatic deletion:

- If customer wants images to be automatically deleted during a defined time period this needs to be enabled["5"] (**not recommended**)

## 6.3 Exclude from Archiving

Note:

These rules only takes effect if automatic Archiving is enabled!

These rules have no impact on the automatic check makes setting in e.g. MR View&GO (when choosing a default archive)

The configuration of "exclude from archiving rules" follows the same logic as the "Autorouting" rules. For details please see chapter 6.4 .

The screenshot shows the 'Exclude from Archive Rules' configuration window. At the top, there's a title bar 'Exclude from Archive Rules' and a sub-header 'Edit / Create Rule'. Below this, there's a 'Condition Name' dropdown menu with 'Select' as the current value, and three buttons: 'New Rule', 'Rename Rule', and 'Delete Rule'. A table with five columns is visible: 'Type', 'Category', 'Operator', 'Value', and 'Delete'. Below the table, there are buttons for 'Add Line', 'Save Rule', and 'Save Rule As'. A yellow warning icon with an exclamation mark is present, followed by the text: 'Excluded objects will be deleted automatically, as soon as they reach a configured age and the Short Term Storage reaches a configured fill level (low watermark). See Automatic Data Deletion Configuration.' Below this, there's a large blue arrow pointing downwards with the text 'Add Rule' inside it. At the bottom, there's a section titled 'Exclusion List' with the subtitle 'Autoexclude from Archiving'. A 'Remove From List' button is located at the bottom left of the window.

## 6.4 Autorouting

### Warning:

In case remote DICOM nodes, used in Autorouting Rules, shall be deleted/renamed, first remove the Autorouting Rule before deleting/renaming the node. Otherwise the rule will not be able to be removed from the list.

### Hint:

In case PACS is not compatible with Phoenix Zip refer to 6.10

The screenshot shows the 'Autorouting Rules' configuration window. It includes a header with 'Edit / Create Rule' and a 'Condition Name' dropdown set to 'Scan Results'. Below this is a table for defining rules with columns for Type, Category, Operator, Value, and a Delete icon. The table contains two rows: 'Dicom Header Attrib' with 'Creator Identifier (0008,0010)' and 'Equals' operator, and 'Dicom Header Attrib' with 'Image Type (0008,0061)' and 'DoesNotContain' operator. Below the table are buttons for 'Add Line', 'Save Rule', and 'Save Rule As'. A 'Choose Routing Target' section shows a list with 'TEAMPLAY', 'TESTPACS', and 'TESTRIS'. An 'Add Rule' button is below this. The 'Autorouting Rule List' section shows a list of rules, with 'Scan Results -> TESTPACS' and 'AV Image Results -> TESTWORKSTATION (default a...' listed. A 'Remove From List' button is below. A 'Save' button is at the bottom. A context menu is open over the rule list, showing options for 'Apply rule' (immediate, during default archiving/send time period) and 'Apply for operation' (Media imported objects, Received and retrieved objects, Saved and sent objects).

**1** Edit / Create Rule

**2** Condition Name: Scan Results

**3** New Rule

**4** Choose Routing Target

**5** Add Rule

**6** Autorouting Rule List

**7** Apply rule

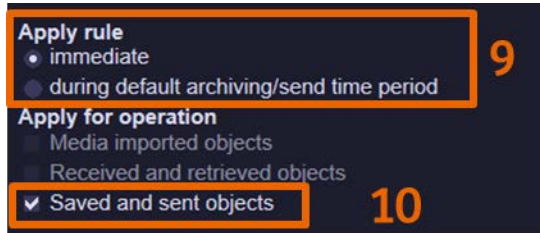
**8** Save

The system comes with 2 preconfigured Autorouting Rules:

- AV Image Results
  - All image results from additional post processing workflows are auto-routed to the target node. (e.g. MR Spectro, MR Cardio, MR Neuro3D)
- Scan Results
  - All scan results, excl. mosaic images, are auto-routed to the target node. Autorouting will occur as soon as scan sequences are completed (not at the end of a scan protocol)

Depending on customer preference these rules need to be configured

1. Choose Rule to be added from drop down list [1] or define New Rule [2]
2. Check Rule and make necessary adjustments [3]
3. Save Rule (otherwise changes will be lost)
4. Choose where the Images using this Rule shall be send to [6]
5. For every Rule make the make the following Checkmarks in [7]:



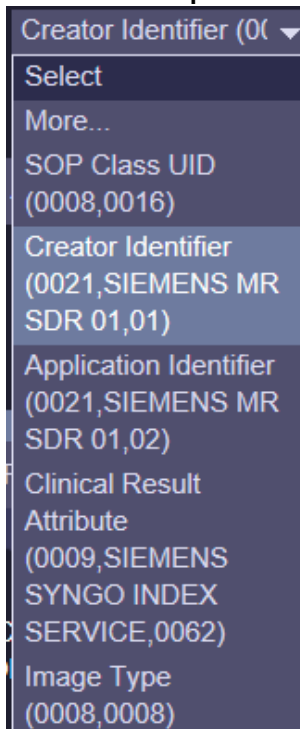
**Hint:**

- “Saved and sent objects” stands for the images created by the MR system (e.g. scan results). The other 2 operation options are relevant for syngo.via workstations, not for a MR scanner (as here no images are uploaded /received by some from some other DICOM node/folder, and edited in some way).
6. Ensure Checkmark to the left of the Rule is set
  7. Press Save [8]

**Hint:**

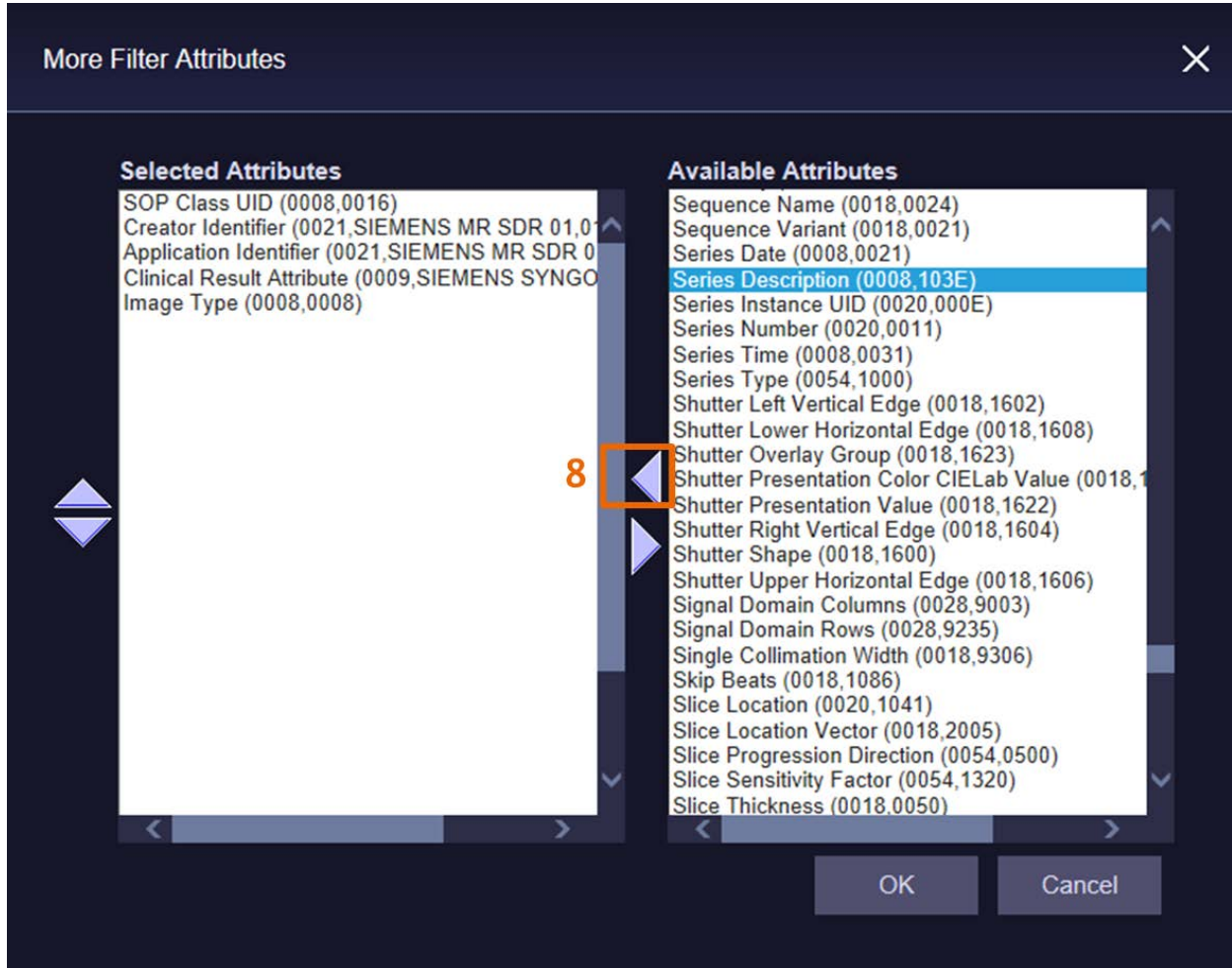
How to add additional Category (e.g. Series Description):

- Check if necessary Category is available from dropdown:





- If the identifier is not available, then choose “More...”. A new Window will appear that shows all available Attributes.



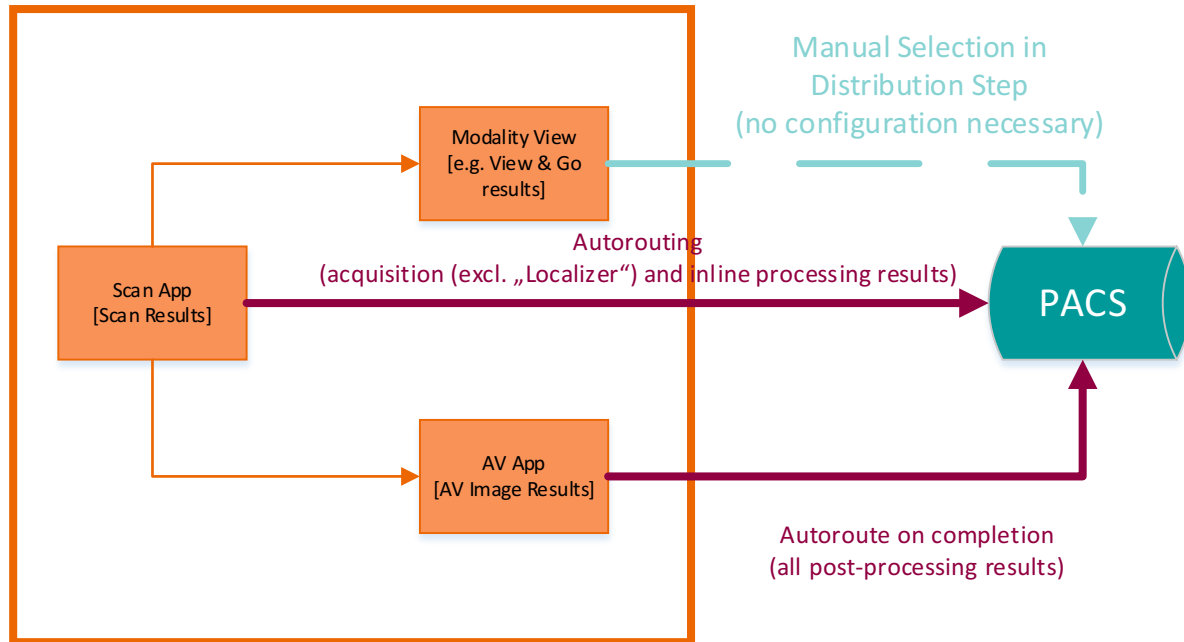
- Choose the necessary attribute and add it to the Selected Attributes by clicking on [8]
- Configure Autorouting Rule accordingly (see Example A for excluding Localizer below).

### 6.4.1 Examples for Autorouting configuration

#### Example A:

- AV Image results to be automatically routed to the PACS after **completed** has been pressed.
- Scanned images (excl. “localizer”) shall be auto-routed to the PACS
- Modality View results will be manually sent (performed by MTA when necessary)
- Individual exclusions from auto-routing can be configured in the protocol properties as in Numaris 4





The Autorouting rules need to be configured as follows:

- **Select Condition Name **Scan Results****
- **Add Line** to Condition
- **Enter the following condition rule**
  - Type: Dicom Header Attributes
  - Category: Series Description
  - Operator: DoesNotContain
  - Value: localizer
- **Save Rule As** enter new condition name (e.g. Scan Results w/o localizer)
- **Select PACS system**
- **Click Add Rule**
  
- **Select AV Image Results**
- **Choose both PACS and Via Workstation as target**
- **Click Add Rule**
  
- **For each Rule in Autorouting Rule List**
  - Select Rule
  - Select Apply Rule: Immediate
  - Select Apply for operation: syno.via created objects
  
- **Test and validate configuration**

**Note:**

Checkmark to the left of the rule needs to be set to activate rule

## Autorouting Rules

Edit / Create Rule

Condition Name **Scan Results w/o localizer** New Rule Rename Rule Delete Rule

|     | Type                   | Category                       | Operator       | Value       | Delete |
|-----|------------------------|--------------------------------|----------------|-------------|--------|
|     | Dicom Header Attribute | Creator Identifier (0008,0010) | Equals         | ACQUISITION |        |
| and | Dicom Header Attribute | Image Type (0008,0002)         | DoesNotContain | MOSAIC      |        |
| and | Dicom Header Attribute | Series Description (0008,0007) | DoesNotContain | localizer   |        |

Add Line

Save Rule Save Rule As

Choose Routing Target

☒ JDPACS (default)

< >

Add Rule

### Autorouting Rule List

Check to set rule as active

- ☒ AV Image Results -> JDPACS (default archive)
- ☒ Scan Results w/o localizer -> JDPACS (default archive)

**Apply rule**

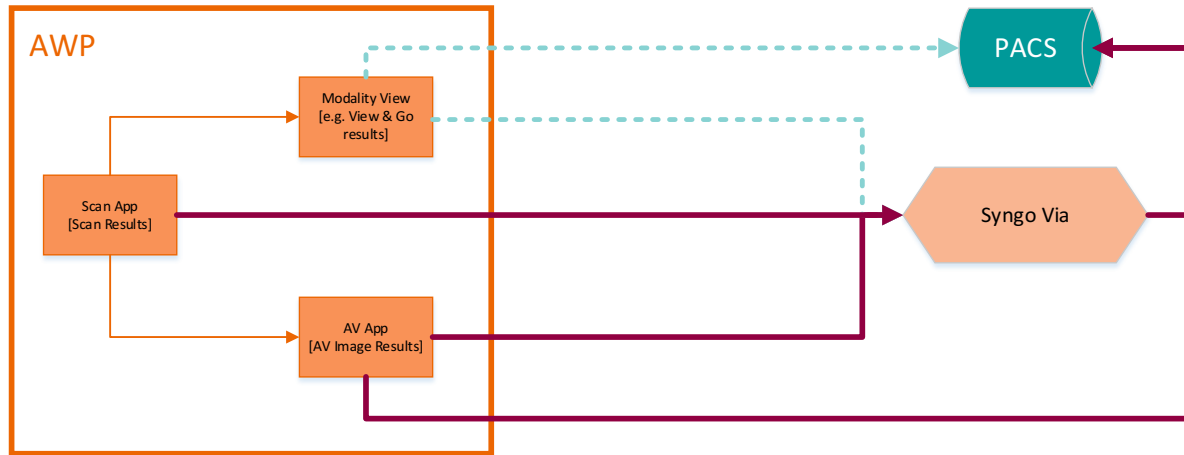
- ☒ immediate
- ☐ during default archiving/send time period

**Apply for operation**

- ☐ Media imported objects
- ☐ Received and retrieved objects
- ☒ syngo.via created objects

### Example B: Complex Setup

- AV Image results to be automatically routed to the PACS **and** syngo Via
- Scanned images shall be automatically routed to the syngo Via
- MR View&GO results shall be manually sent to the syngo Via and PACS
- Syngo.Via shall automatically route results to the PACS (needs to be set up in syngo Via Workstation)



--- Manual Transfer  
— Automatic Transfer

- Select **Scan Results**
- Choose syngo.Via as Target
- Click **Add Rule**
- Select **AV Image Results**
- Choose both **PACS and syngo.Via** as target
- Click **Add Rule**
- For **each** Autorouting Rule in Rule List
  - Select Rule
  - Select Apply Rule: Immediate
  - Select Apply for operation: syngo.via created objects
- Configure syngo.Via to automatically route results to PACS
- Test and validate Configuration

### Note:

Checkmark to the left of the rule needs to be set to activate rule

**SIEMENS** NX201703171M / medadmin Basic (Restricted Remote Access)

Diagnose Repair Installation Technical Configuration Quality Assurance Clinical Configuration

▶ Data Preprocessing  
▶ Patient Browser  
Prior Study Search  
▶ Reading  
DICOM Data Handling  
Archiving and Deletion  
Exclude from Archive  
**Autorouting**  
Job Settings  
Media Compression  
Export Path  
MPPS Completion  
DICOM Study Description  
▶ Patient Browser - Scheduler  
Configuration Export/Import  
▶ Licensing  
Layout Management  
User and Role Administration  
Expert-i

## Autorouting Rules

Edit / Create Rule

Condition Name: AV Image Results New Rule Rename Rule Delete Rule

| Type                    | Category               | Operator       | Value                 | Delete |
|-------------------------|------------------------|----------------|-----------------------|--------|
| Dicom Header Attrib     | Creator Identifier (00 | Equals         | APPLICATION           |        |
| and Dicom Header Attrib | SOP Class UID (00      | DoesNotContain | 1.2.840.10008.5.1.4.1 |        |
| and Dicom Header Attrib | SOP Class UID (00      | DoesNotContain | 1.2.840.10008.5.1.4.1 |        |
| and Dicom Header Attrib | SOP Class UID (00      | DoesNotContain | 1.2.840.10008.5.1.4.1 |        |
| and Dicom Header Attrib | SOP Class UID (00      | DoesNotContain | 1.2.840.10008.5.1.4.1 |        |

Add Line Save Rule Save Rule As

**Choose Routing Target**  
☒ PACS (default)  
☐ VIA

Add Rule

**Autorouting Rule List**  
 Check to set rule as active  
☒ Scan Results -> VIA  
☒ AV Image Results -> PACS (default archive),VIA

**Apply rule**  
☒ immediate  
☐ during default archiving/send time period

**Apply for operation**  
☐ Media imported objects  
☐ Received and retrieved objects  
☒ syngo via created objects

## 6.4.2 Validating Autorouting

- Perform scan / complete post processing to trigger Autorouting.
- Verify if Autorouting has been successfully completed (Patient Browser and Remote Node)

## 6.5 Job Settings

## Automatic Retry for Network Jobs

|                                              |        |
|----------------------------------------------|--------|
| Number of retries for network transfer jobs  | 2      |
| Delay of first retry                         | 30 sec |
| Multiplication factor for subsequent retries | 2.0    |

No configuration necessary

## 6.6 Media Compression

### Default Media Compression

**i** The system will first try to use the compression algorithm with the highest priority. If this compression algorithm cannot be applied, the system will try to use the algorithm with the next lowest priority.

**Choose Compression Algorithm**

|            |                                                                        |
|------------|------------------------------------------------------------------------|
| Priority 1 | JPEG Lossless, Non-Hierarchical, First-Order Prediction (Process 14) ▼ |
| Priority 2 | JPEG 2000 Image Compression (Lossless Only) ▼                          |
| Priority 3 | Not Set ▼                                                              |

No configuration necessary

## 6.7 Export Path

- Enter the preferred Export Path into **"Path"** and enter a name into **Description** (the name in the Description will be shown as choices in the Export menu)

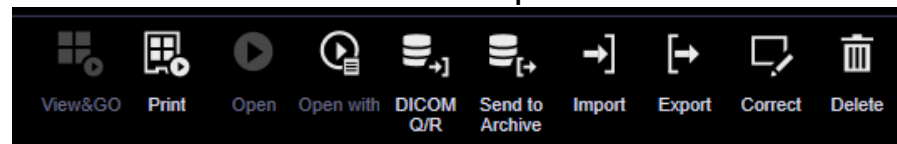
### DICOM Export Path

| Path                                                                                           | Description |
|------------------------------------------------------------------------------------------------|-------------|
| <div><b>i</b> Please enter a valid UNC (e.g. "\\servername\pathname\...") or local path.</div> |             |

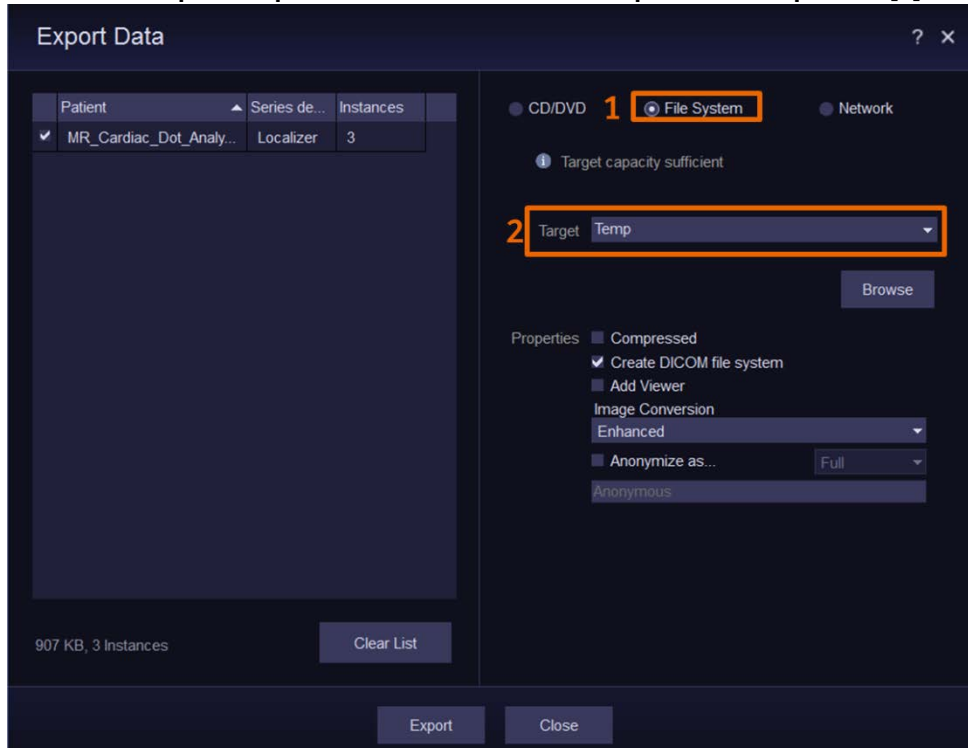
[Add New Path](#)

[Save](#)

- In the Patient Browser click on export



- Click on **File System** [1]
- The defined Export Path can now be chosen from the drop down [2]



**Warning:** Depending on the software version the export dialogue list is only cleared after pressing export. (i.e. export a data set, close the dialogue and export the next patient can lead to two different patients in the export list when the dialogue pops-up again)

**Hint:**

- When choosing **Add Viewer** the image conversion is automatically changed to *interoperability*
- When exporting to integrated DVD Burner.
  - MTRA needs to choose a Profile (media burning profile). Depending on the Profile, the Viewer is automatically added, interoperability activated and anonymization activated.
  - The Media Burning Profiles can be configured in **Configuration Panel → Export**
- Exported DICOM Viewer is the same standard viewer as with VD SW-Line (Numaris/4).
- When exporting to any Network Node, the exporting rules are set according to the configuration in the Remote DICOM node.

Set properties according to customer preference:

- **Compressed:** Images are compressed to make optimum use of the available storage space (for settings see 6.6)
- **Create DICOM file system:** DICOM DIR will also be generated. Filename and Structure will change.
- **Add Viewer:** This adds a DICOM viewer to the export location, with this it is possible to view the DICOM images without the need of any additional software.
- In **Image Conversion** it is possible to choose if the export is in enhanced DICOM or standard (*Interoperability*). By standard it is set to *Enhanced*.
- In **Anonymize as...** it is possible to anonymize the patient information for exporting.

- *Full*: DICOM-conform anonymization
- *Reduced*: DICOM-conform anonymization. This option keeps UIDs, patient characteristics and date.
- *Service*: Anonymization for service purposes.

### 6.7.1 Validation Export Data

- Export test DICOM Images to destination
- Check destination for exported DICOM Images

## 6.8 MPPS Completion

To configure MPPS also the steps described in 5.2.8 necessary

**MPPS Completion**

Mode **Automatic** ▼

**Automatic**

- MPPS will be set to status COMPLETED and sent to the RIS when Patient is closed in the Examination or the next Patient is loaded in Examination

**Manual**

- Manual control of the MPPS status by the user

**Save**

This setting specifies when the MPPS completion will be send:

- **Automatic**: the MPPS will be automatically completed when Patient is closed or next Patient loaded to examination (**recommended**)
- **Manually**: the user has to manually set the completion in the Patient Browser (Scheduler)

## 6.8.1 Validation MPPS

See 5.2.8.1 for details

## 6.9 DICOM Study Description

The default setting for the Study Description can be configured to be set with the following values:

- **Modality:** Region^Examination
  - **Modality including Program level:** Region^Examination^Program
  - **Requested Procedure:** Scheduled by RIS, DICOM Worklist (0023,1060)
  - **Scheduled Procedure Step Description:** Scheduled by RIS, DICOM Worklist (0040,0007)
- Discuss with customer which setting is desired. Default setting is **Modality**.

The screenshot shows a configuration window titled "DICOM Study Description". At the top, there is a label "Study Description" followed by a dropdown menu currently showing "Modality". Below this, the window is organized into sections with their respective default values and notes:

- Modality**
  - Dot Cockpit: Region^Examination
- Modality including Program level**
  - Dot Cockpit: Region^Examination^Program
  - Could enhance syngo.via workflow mapping accuracy.
- Requested Procedure**
  - Scheduled by RIS, DICOM Modality Worklist (0032,1060)
- Scheduled Procedure Step Description**
  - Scheduled by RIS, DICOM Modality Worklist (0040,0007)

At the bottom left of the window is a "Save" button.



## 6.10 Phoenix Zip

### Note:

- This setting does not affect manual sending from Patient Browser.
- Changes only affect newly scanned images
  - If historic Images are sent via the View&GO Distribution Step, the Phoenix ZIP preselection is not affected by the change.

With this setting Phoenix ZIP sending can be disabled for Autorouting and it deselects the “Default Archive” destination in View&GO distribution Step. So this option synchronizes Autorouting and Distribution Step behavior. Use this option if customer does not want Phoenix ZIP at PACS or PACS is incompatible with these objects.

In case Phoenix ZIP is not to be send to the PACS system, disable this here.

## PhoenixZIP

Automatic Sending **Enabled** 1

Enabled

- PhoenixZip is included in Autorouting
- The "Default Archive" target is preselected in the View&GO Distribution Step UI

Disabled

- PhoenixZip is excluded from Autorouting
- The "Default Archive" target is not selected in the View&GO Distribution Step UI

Manual sending is not affected by this setting at all.

**Save** 2

## 7 How to verify correct transfer of DICOM image

The following describes how to verify a correct setup of the DICOM nodes and DICOM Data Handling

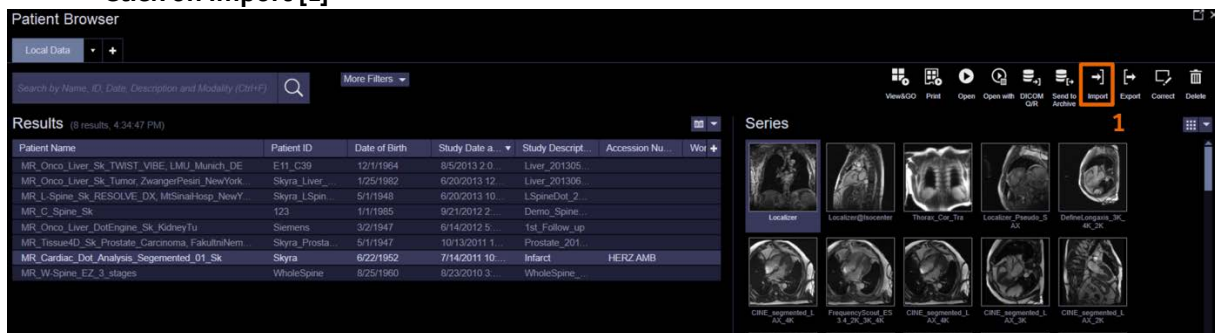
### 7.1.1 Importing Test images

Test images to verify a correct DICOM image transfer can be found in the following folder:

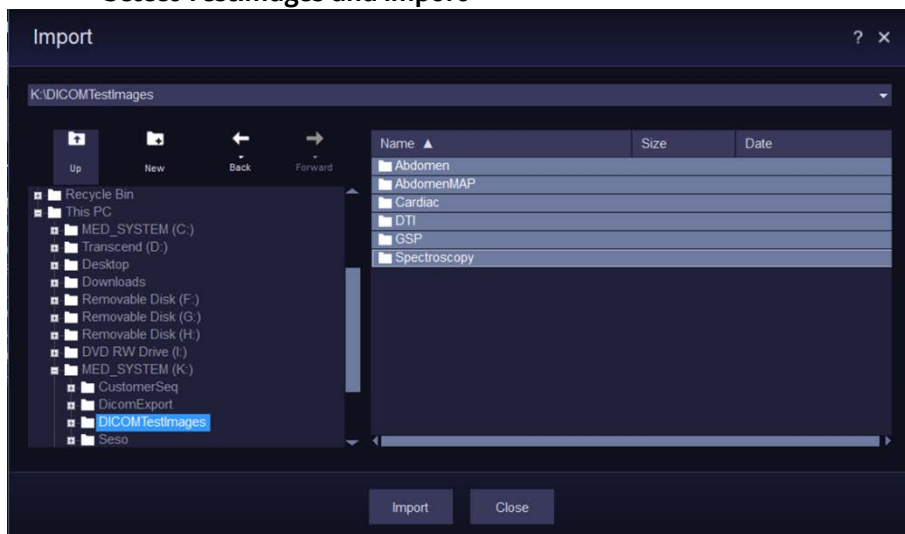
**K:/DICOMTestImages**

To import these DICOM files into the Patient Browser:

- Click on Import [1]



- Select Testimages and import



## The Test images will be loaded to the patient Browser

**Patient Browser**

Local Data

Search by Name, ID, Date, Description and Modality (Ctrl+F)

More Filters

**Results** (14 results, 4:38:54 PM)

| Patient Name                                | Patient ID       | Date of Birth | Study Date and Time | Study Description  | Accession Number | Worklist |
|---------------------------------------------|------------------|---------------|---------------------|--------------------|------------------|----------|
| SiemensTest_Spectroscopy                    | 3333-SMR-7       | 4/24/1971     | 5/18/2017 11:00     | breast Breast...   |                  |          |
| SiemensTest_GSP_Graphic                     | 2017.03.06-1     | 3/6/1973      | 3/6/2017 12:00      | head clinical I... |                  |          |
| SiemensTest_DTI                             | 2017.05.18-1     | 1/1/1947      | 5/18/2017 10:00     | head clinical      |                  |          |
| SiemensTest_Cine_CS_3h                      | SP               | 6/6/1957      | 3/3/2017 11:00      | heart Cardiac      |                  |          |
| SiemensTest_AbdomenMaps                     | 123456-SMR       | 7/2/1973      | 5/18/2017 3:00      | whole body o...    |                  |          |
| SiemensTest_Abdomen                         | 2017.05.18-0     | 1/1/1967      | 5/18/2017 9:00      | Abdomen Ab...      |                  |          |
| MR_T1_Spine_Sk_Skylar                       | MR_T1_Spine      | 6/22/1952     | 6/22/2013 2:00      | T1_Spine_Sk...     |                  |          |
| MR_T1_Spine_Sk_Phottal_Carcinoma_FatSatNem  | Skylar_Phota...  | 5/1/1947      | 10/13/2011 1:00     | Prostate_201...    |                  |          |
| MR_Orco_Liver_Sk_TWIST_VIBE_LM1_Munich_DE   | E11_C99          | 12/1/1964     | 8/5/2013 7:00       | Liver_201306       |                  |          |
| MR_Orco_Liver_Sk_Tumor_ZwangerPeest_NewYork | Skylar_Liver_... | 1/25/1982     | 6/20/2013 12:00     | Liver_201306       |                  |          |
| MR_Orco_Liver_DoEngine_Sk_KidneyTu          | Siemens          | 3/2/1947      | 6/14/2012 5:00      | T1_Follow_up       |                  |          |
| MR_L_Spine_Sk_RESOLVE_DX_MSirolHosp_NewY    | Skylar_L_Spin... | 5/1/1948      | 6/22/2013 10:00     | L_SpineDot_2...    |                  |          |
| MR_Cardiac_Dot_Analys_Segmented_01_Sk       | Skylar           | 6/22/1952     | 7/14/2011 10:00     | Infant             | HERZ AMB         |          |
| MR_C_Spine_Sk                               | 123              | 1/1/1985      | 9/21/2012 2:00      | Demo_Spine...      |                  |          |

**Series**

The image thumbnails are arranged in a 3x4 grid. The first row shows: a brain scan (axial), a brain scan (sagittal), a brain scan (coronal), a brain scan (axial), a brain scan (sagittal), and a brain scan (coronal). The second row shows: a brain scan (axial), a brain scan (sagittal), a brain scan (coronal), a brain scan (axial), a brain scan (sagittal), and a brain scan (coronal). The third row shows: a brain scan (axial), a brain scan (sagittal), a brain scan (coronal), a brain scan (axial), a brain scan (sagittal), and a brain scan (coronal).

## 7.1.2 Verifying correct DICOM setup

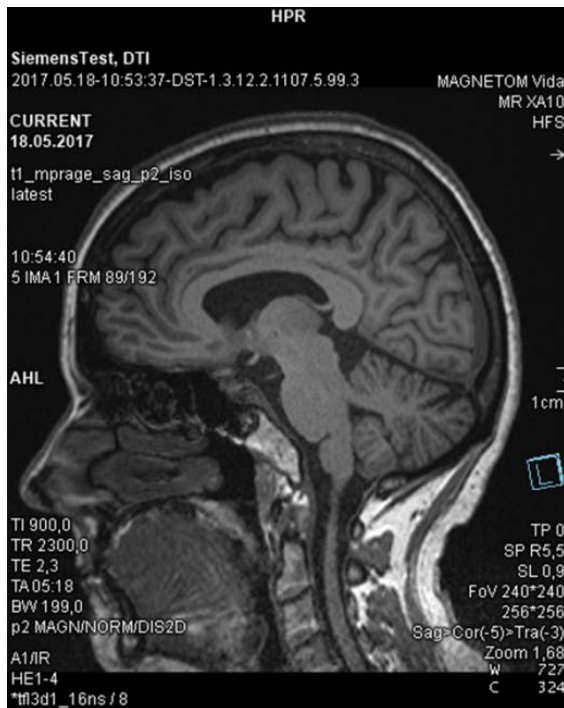
- Send the imported test images with export to the respective remote node
- Verify DICOM images on the remote node

### 7.1.2.1 Common Enhanced MR Image

Images to be used:

- SiemensTest, Abdomen > Series 32, 33, 34, 37, 40, 41, 45, 50
- SiemensTest, AbdomenMaps > Series 53, 82, 83
- SiemensTest, Cine\_CS\_bh > Series 8,9, 38
- SiemensTest, DTI > Series 5, 23

Please Verify:



- Navigation through the frames
- Window Values applied
- Image Text fulfills customer needs, especially MR parameter

Solution:

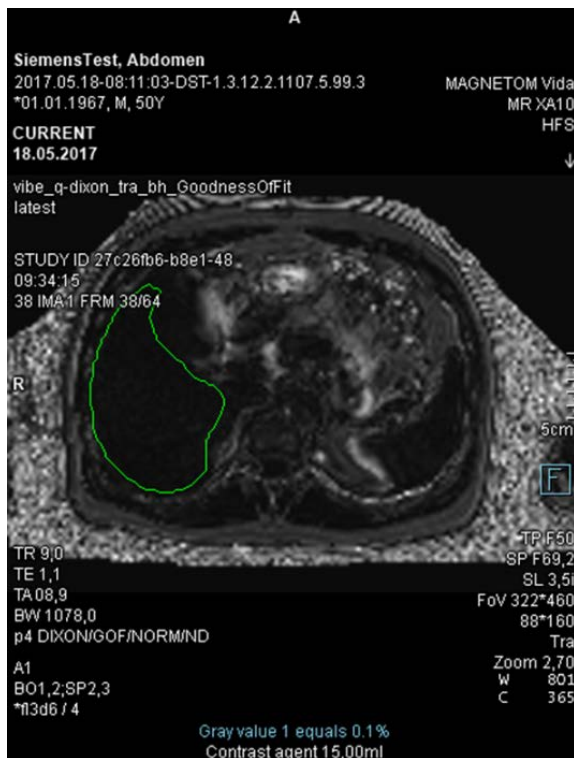
- Convert Enhanced MR and related GSPS

### 7.1.2.2 Enhanced MR Image + Presentation State Graphics

Images to be used:

- SiemensTest^Abdomen > Series 37, 38, 39, 42, 43, 44 (Liver Segmentation)
- SiemensTest^Cine\_CS\_bh > Series 38, 39 (Inline VF)

Please Verify:



- See common Enhanced MR Image
- Graphics applied at least without specified color (many systems don't evaluate the color of the graphic elements)

Solution:

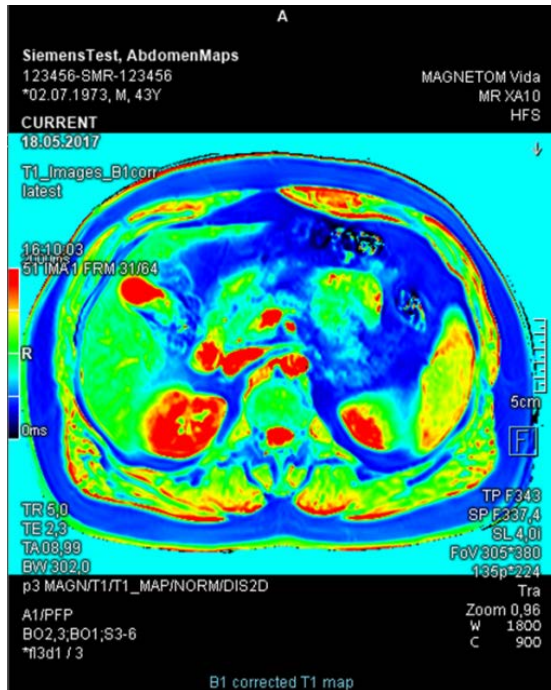
- Convert Enhanced MR and related GSPS

### 7.1.2.3 Enhanced MR Image with Color Lookup Table

Images to be used:

- SiemensTest^AbdomenMaps > Series 51, 54 (52, 55 PR)

Please Verify:



- See common Enhanced MR Image
- Color LUT applied
- (Note: Usually also a Presentation State is created for Color Bar annotations)

Solution:

- Convert Enhanced MR Color LUT and related GSPS

### 7.1.2.4 Enhanced MR Image with Image and Frame comments

Images to be used:

- SiemensTest^Cine\_CS\_bh > Series 8

Please Verify:



- Navigation through the frames
- Window Values applied
- Image Text fulfills customer needs, especially MR parameter  
→ special use case with comments on instance and frame level (if important for the customer)

Solution:

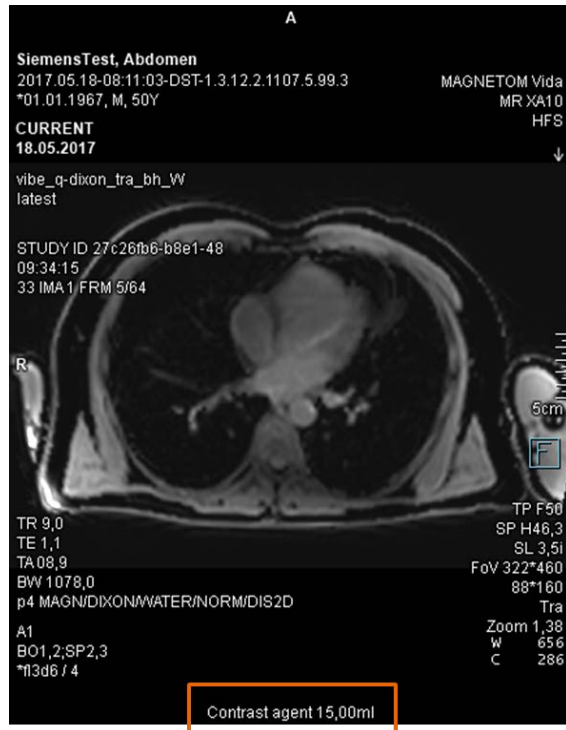
- Convert Enhanced MR and related GSPS

### 7.1.2.5 Enhanced MR Image with Contrast Bolus Information

Images to be used:

- SiemensTest^Abdomen > Series 33

Please Verify:



- Navigation through the frames
- Window Values applied
- Image Text fulfills customer needs, especially MR parameter  
→ special use case with Contrast Bolus information (if important for the customer)

Solution:

- Convert Enhanced MR and related GSPS

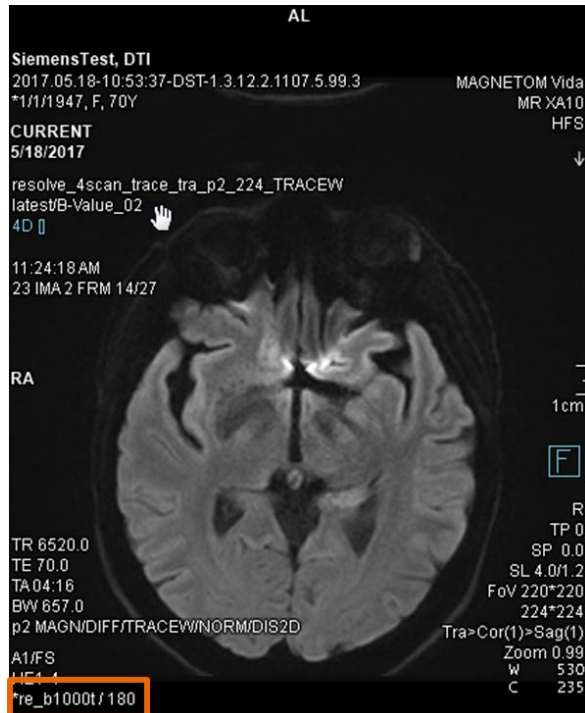


### 7.1.2.6 Enhanced MR Image with Diffusion B-Values

Images to be used:

- SiemensTest^DTI > Series 23 (B0/B1000)

Please Verify:



- Navigation through the frames
- Window Values applied
- Image Text fulfills customer needs, especially MR parameter  
→ special use case with Diffusion B-Values (if important for the customer)

Solution:

- Convert Enhanced MR and related GSPS

## 7.1.2.7 Enhanced MR Image 3D/4D

Images to be used:

- SiemensTest^DTI > Series 5 (3D)
- SiemensTest^AbdomenMaps > Series 82 (4D)

Please Verify:

|    |                       |    |                |
|----|-----------------------|----|----------------|
| 82 | ep2d_perf_p2_MoCo     | MR | 5/18/2017 4... |
| 83 | TTP_ep2d_perf_p2_MoCo | MR | 5/18/2017 4... |

stances

| ▲  | Acquisition Date an... | Image Comments        | Number of Fra... | SOP Class UID |
|----|------------------------|-----------------------|------------------|---------------|
| 1  | 5/18/2017 4:40:11...   | Reference volume...   | 22               | MR enh        |
| 2  | 5/18/2017 4:40:13...   | Motion: 0.11,0.00...  | 22               | MR enh        |
| 3  | 5/18/2017 4:40:14...   | Motion: 0.00,0.00...  | 22               | MR enh        |
| 4  | 5/18/2017 4:40:16...   | Motion: 0.11,0.00...  | 22               | MR enh        |
| 5  | 5/18/2017 4:40:17...   | Motion: 0.00,0.00...  | 22               | MR enh        |
| 6  | 5/18/2017 4:40:19...   | Motion: 0.00,0.00...  | 22               | MR enh        |
| 7  | 5/18/2017 4:40:20...   | Motion: 0.11,0.00...  | 22               | MR enh        |
| 8  | 5/18/2017 4:40:22...   | Motion: 0.00,0.00...  | 22               | MR enh        |
| 9  | 5/18/2017 4:40:23...   | Motion: 0.11,0.00...  | 22               | MR enh        |
| 10 | 5/18/2017 4:40:25...   | Motion: 0.00,0.00...  | 22               | MR enh        |
| 11 | 5/18/2017 4:40:26...   | Motion: -0.11,0.00... | 22               | MR enh        |
| 12 | 5/18/2017 4:40:28...   | Motion: 0.00,0.00...  | 22               | MR enh        |
| 13 | 5/18/2017 4:40:29...   | Motion: 0.00,0.00...  | 22               | MR enh        |
| 14 | 5/18/2017 4:40:31...   | Motion: 0.11,0.00...  | 22               | MR enh        |
| 15 | 5/18/2017 4:40:32...   | Motion: 0.00,0.00...  | 22               | MR enh        |
| 16 | 5/18/2017 4:40:34...   | Motion: 0.11,0.00...  | 22               | MR enh        |
| 17 | 5/18/2017 4:40:35...   | Motion: 0.11,0.00...  | 22               | MR enh        |
| 18 | 5/18/2017 4:40:37...   | Motion: 0.00,0.00...  | 22               | MR enh        |
| 19 | 5/18/2017 4:40:38...   | Motion: 0.00,0.00...  | 22               | MR enh        |
| 20 | 5/18/2017 4:40:40...   | Motion: 0.11,0.00...  | 22               | MR enh        |
| 21 | 5/18/2017 4:40:41...   | Motion: 0.00,0.00...  | 22               | MR enh        |
| 22 | 5/18/2017 4:40:43...   | Motion: 0.00,0.00...  | 22               | MR enh        |
| 23 | 5/18/2017 4:40:44...   | Motion: 0.11,0.00...  | 22               | MR enh        |

- For checking 3D functionality (if needed)
- For checking multiple Stacks within one Series
  - 4D navigation at PACS
  - Separate Series options

Solution:

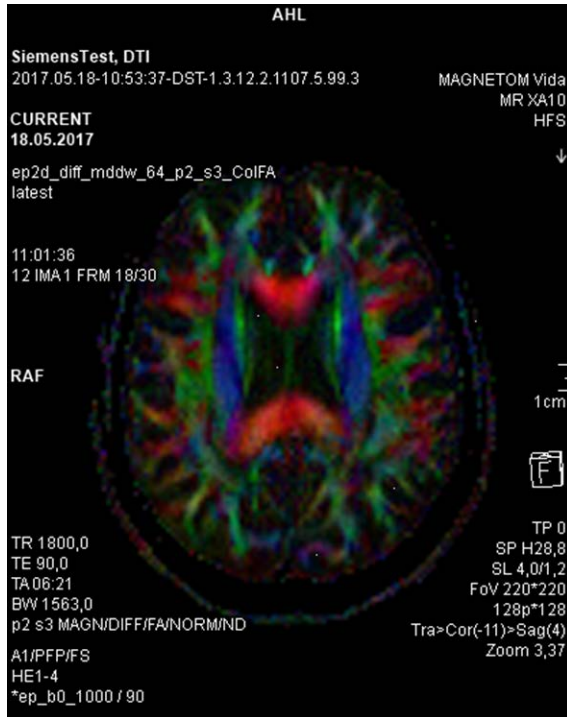
- Convert Enhanced MR and related GSPS
- Separate Series for each converted Multiframe Image

### 7.1.2.8 Enhanced MR Color Image

Images to be used:

- SiemensTest^DTI > Series 12  
(DTI Color FA)

Please Verify:



- Navigation through the frames
- Color displayed
- Image Text fulfills customer needs, especially MR parameter

Solution:

- Convert Enhanced MR Color

### 7.1.2.9 Raw Data (DTI Tensor)

Images to be used:

- SiemensTest^DTI > Series 13 (DTI Tensor)

Please Verify:

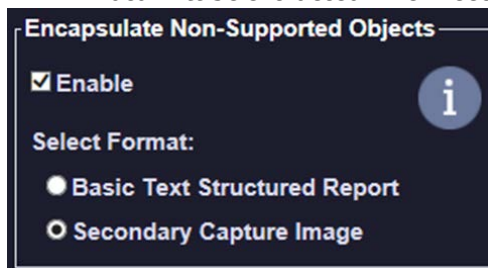
| Number ▲      | Series Description                     | Modality       |
|---------------|----------------------------------------|----------------|
| 5             | t1_mprage_sag_p2_iso                   | MR             |
| 12            | ep2d_diff_mddw_64_p2_s3_ColFA          | MR             |
| 13            | ep2d_diff_mddw_64_p2_s3_TENSOR         | MR             |
| 23            | resolve_4scan_trace_tra_p2_224_TRAC... | MR             |
| ces           |                                        |                |
| SOP Class UID | Acquisition Date a...                  | Image Comments |
| RAW           | 17.05.2017 10:47:...                   |                |

- DICOM Send and Receive successful

Solution:

- Encapsulate None-Supported Objects

→ Data will be extracted when received by Numaris/X or syngo.via



## 7.1.2.10 MR Spectroscopy

Images to be used:

SiemensTest^Spectroscopy > Series 12  
(Spectroscopy data)

Please Verify:

| Series Number ▲ | Series Description | Modality |
|-----------------|--------------------|----------|
| 12              | svs_se_breast      | MR       |

ances

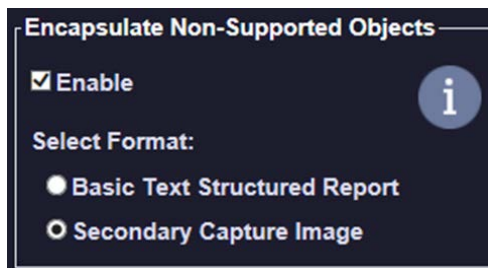
| ▲ | SOP Class UID | Acquisition Date a... | Image Comment |
|---|---------------|-----------------------|---------------|
| 1 | MR spect      | 18.05.2017 13:15:...  | _pwc_18       |

- DICOM Send and Receive successful

Solution:

- Encapsulate None-Supported Objects

→ Data will be extracted when received by Numaris/X or syngo.via

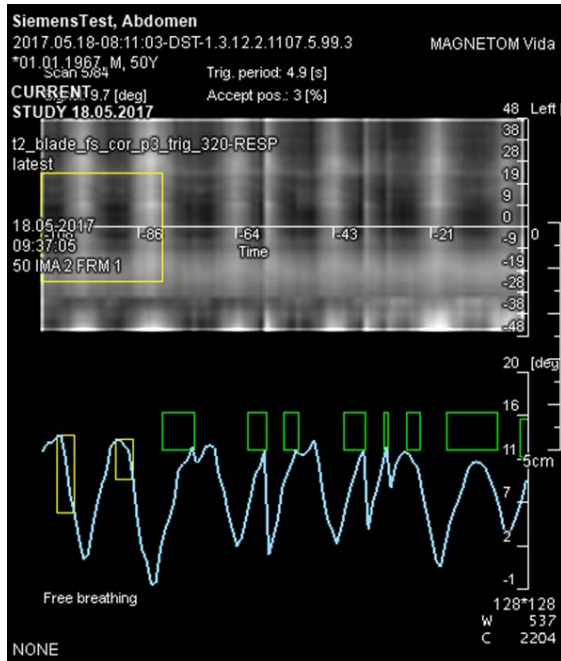


### 7.1.2.11 Multi-Frame Grayscale Word Secondary Capture & Presentation State Graphics (Inline Reports)

Images to be used:

- SiemensTest^Abdomen > Series 35, 36, 48, 49, 50, 51
- SiemensTest^AbdomenMaps > Series 41, 42
- SiemensTest^Cine\_CS\_bh > Series 40

Please Verify:

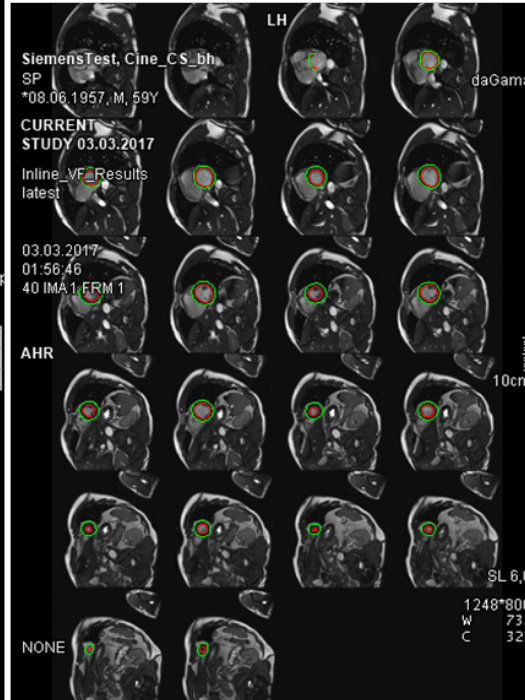
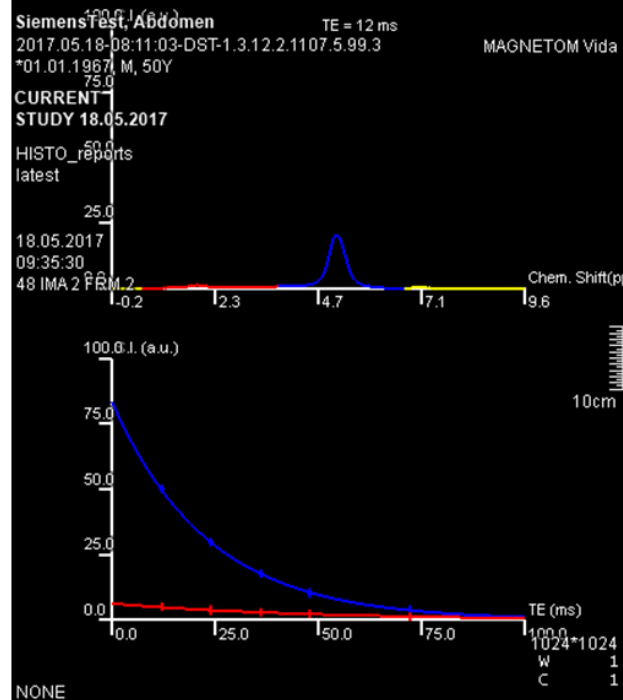
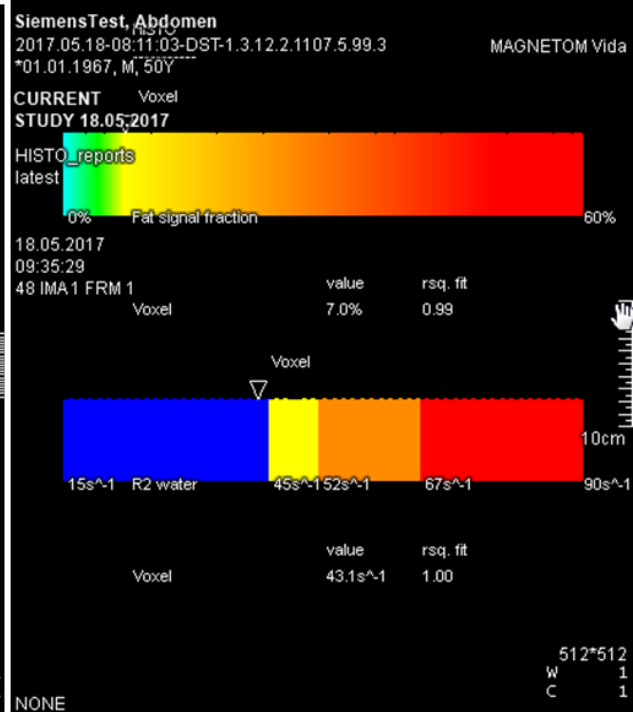
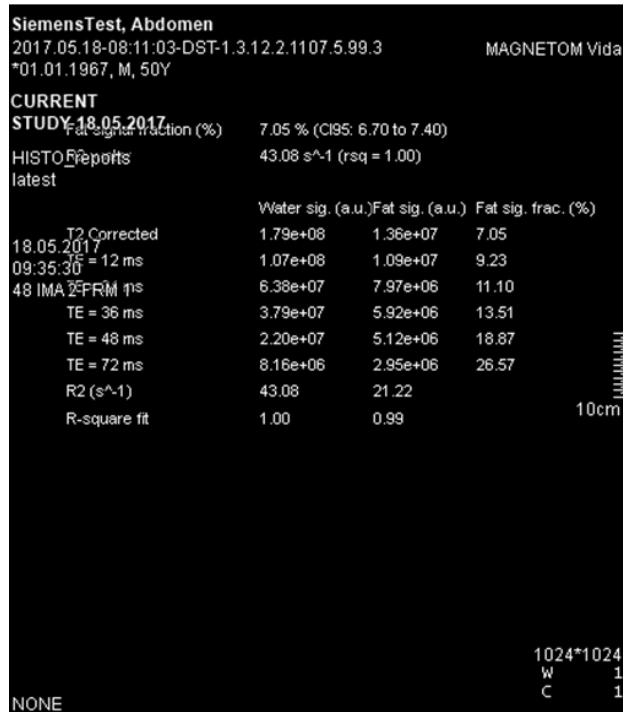


- Graphics and Text readable, position is consistent
- Many systems don't evaluate the color of the graphic elements in the PR. Usually Color is needed here for correct interpretation

Solution:

- Convert Multiframe Grayscale Word SC and related GSPS

Further Example images:



## 8 Troubleshooting

To analyze problems during the DICOM transfer / receive, it may be necessary to trace the DICOM transfers. The following describes the steps necessary to perform this trace.

**Hint:** How to enter the file explorer (exit kiosk mode)

- Press the following keys: **Tab + Delete + [→]** (key is located on the most right of the num pad)
- Enter User: **medadmin**
- Enter password: (contact Siemens Service for support)
- By taping the windows key on the keyboard you can now enter the start menu / file explorer

### 8.1.1 Starting DICOM Trace

#### 1. Prepare / create folder to capture trace data

Recommended C:\temp\trace (these files will be deleted after a back-up)

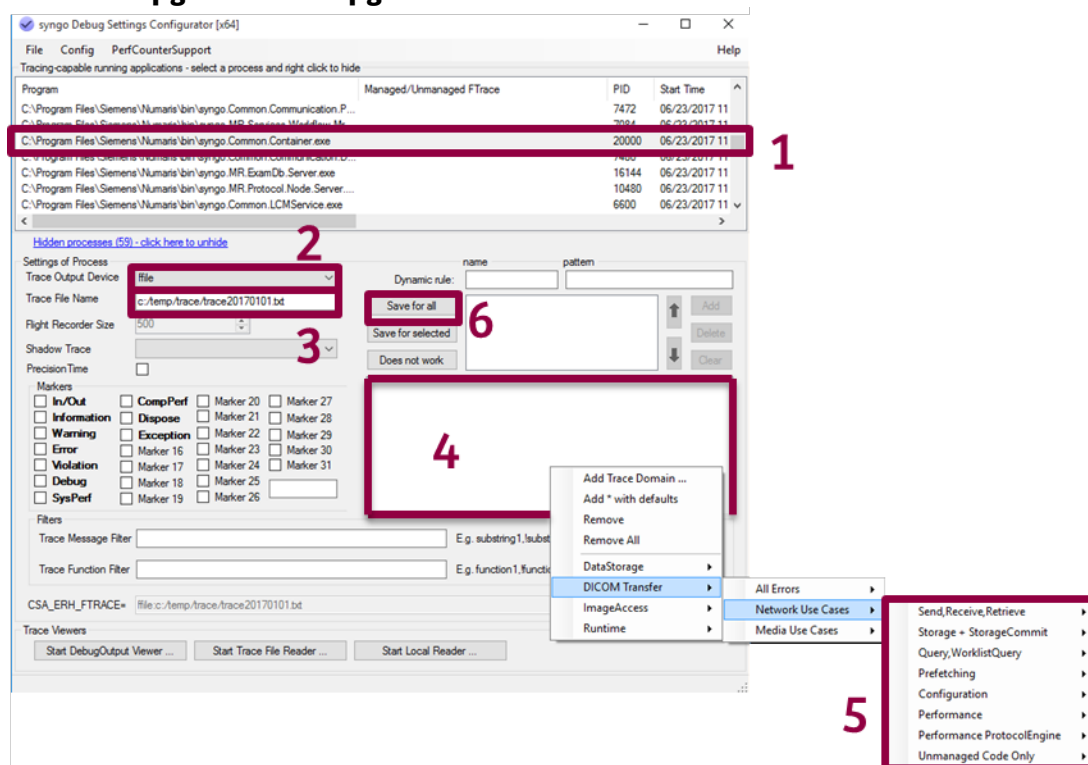
#### 2. Workplace (MRAWP)

The task to be captured should start as soon as possible once the trace is enabled. This will shorten the time when evaluating the trace files (open client; select data to be send)

#### 3. Start trace configurator

Start the syngo.DebugConfig.exe in admin mode (right click on file → run as administrator)  
C:\Program Files\Siemens\Numaris\bin\syngo.DebugConfig.exe

#### 4. Configure trace configurator





- Select C:\Program Files\ Siemens\Numaris\bin\ **syngo.Common.Container.exe** [1]
- Set Trace Output Device to **ffile** [2]
- Set trace file name to capture folder and set a name  
e.g. C:\Temp\Trace\trace20170101.txt
- Right click in frame [4]
- Select wanted trace profile from **DICOM Transfer → Network Use Cases →**
  - **Send, Receive, Retrieve**
  - **Prefetching**
  - **Query, Worklist Query**
- In the wanted trace profile choose → **Verbose (+Debug)**  
(The settings will now be displayed in frame [4])
- Start the trace by clicking **Save for all** [6]
- Perform task that causes problems
- Stop the trace by clicking setting the **trace output device** to **off** [2]  
and then clicking **Save for all** [6]

### 8.1.2 Transferring DICOM Trace Data and clean up

- Transfer Trace Data to RSC / HSC
- Delete the trace output folder

## 9 Enhanced DICOM why and what is this

“Enhanced” is not a MR-only topic. It is also available for CT and other applications. It is a standard since 2002, and since then has been a common principle for other modalities: Enhanced CT, Enhanced US, Enhanced XA, Enhanced XRF, Enhanced PET.

Knowing that not all customers are able to handle this standard or have issues with part of this, it is possible to convert the data back to the old standard in the Export rules of every remote DICOM node (see 5.2.5 for details)

### 9.1 DICOM objects Numaris/x Vs Numaris/4

The following table describes the changes in the DICOM object types for Numaris/4 to Numaris/X. The reason for the change is:

- Up to date application support and standardized graphics
- Replacement of private objects with standard DICOM objects (→ archivable)

| Numaris/4                                    | Numaris/X                                          |
|----------------------------------------------|----------------------------------------------------|
| MR Image, Mosaic                             | Enhanced MR                                        |
| Secondary Capture Image                      | Enhanced MR Color                                  |
| CsaNonImage (Spectroscopy)                   | MR Spectroscopy                                    |
| CsaNonImage (Tensor)                         | Raw Data                                           |
| syngo Object Oriented Graphics/DICOM Overlay | Grayscale Softcopy Presentation State              |
| Structured Reports (PhoenixZIP)              | Structured Reports (PhoenixZIP)                    |
| Secondary Capture Image                      | Multi-frame Grayscale Word Secondary Capture Image |

### 9.2 What is different on enhanced MR

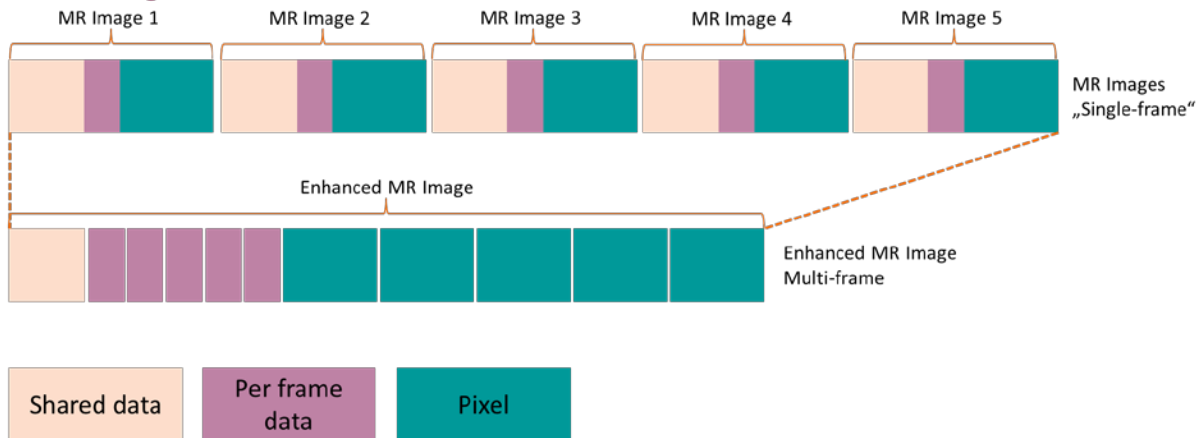
The following gives some details on what exactly has changed on the DICOM formatting, to get a feeling for the effect when converting back the images during export (see 5.2.5).

#### 9.2.1 Single-frame Vs. Multiframe structure

Moving from “Single-Frame” to a Multi-frame structure results in:

- More efficient memory structure
- Faster read/write
- Less file input / Output
- Unit to handle: slice group/volume, no single slice

### From “Single-frame” to Multi-frame structure



## 9.2.2 Memory footprint, Performance, Organization structure

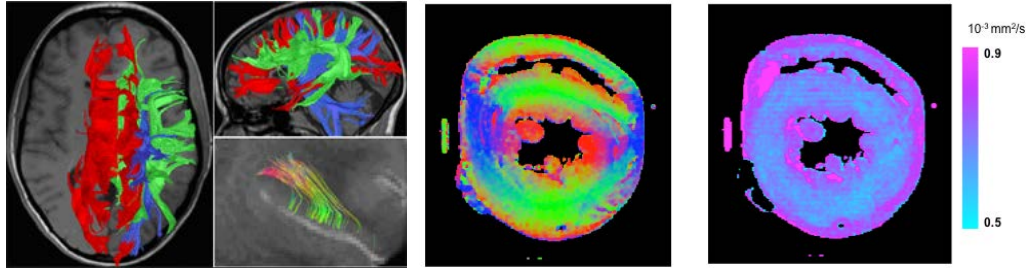
- Lower memory footprint, more efficient structure
  - Phoenix protocol data only once per Multi-frame needed!
- Navigation capabilities:
  - Dimension Index, Stacks, Position index, Temporal Position Index
  - Supports correct sorting and display
- Better performance DICOM Networking, Export, Loading

## 9.2.3 Attributes: much more, strict, and structured

- More and strict definitions
- Get rid of the optional attributes
- Extended MR Parameter (timing, acquisition techniques, coils, ...)
- Application support
  - Diffusion
  - ASL
  - Cardiac
  - Color Maps
  - Real World Value Mapping

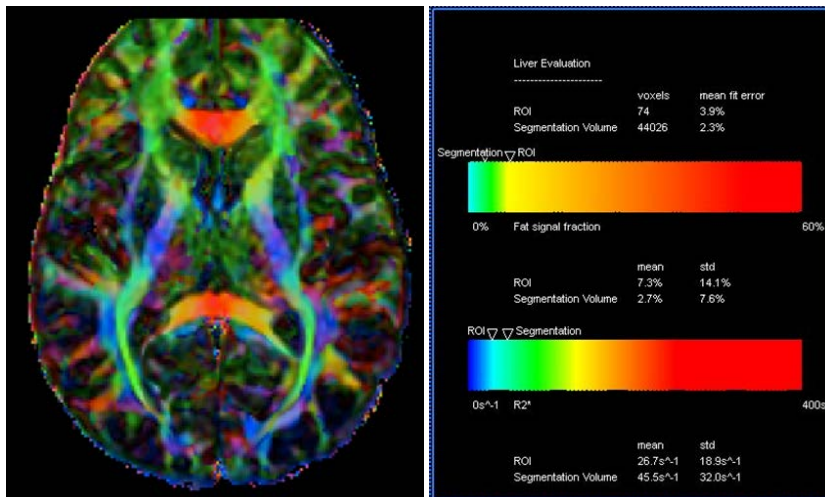
## 9.2.4 Color support

- Supplemental Palette Color Lookup Table
- Real World Value Mapping
- Color Pixel



## 9.2.5 Graphics

High quality standard graphics with Presentation States



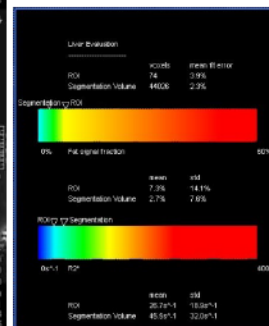
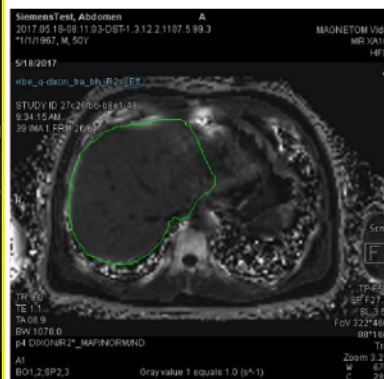
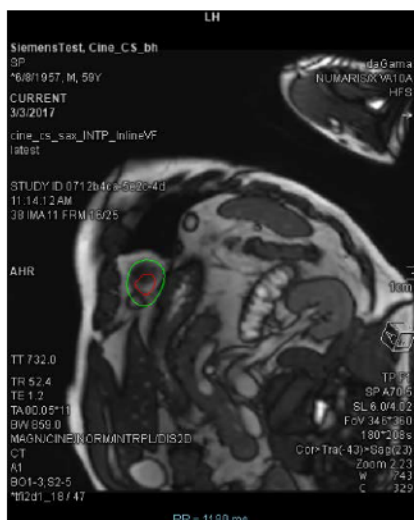
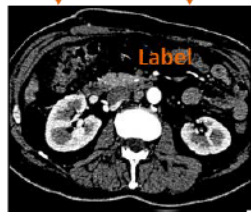
## 9.2.6 Grayscale Softcopy Presentation State (PR or GSPS)

GSPS is a separate DICOM object that contains information on how a particular image should be displayed.

- Windowing, zoom, panning, rotations, ...
- Graphics (Label, annotation, line, circle, ...)
  - ☐ high quality graphics
- PR is referencing the Image
  - only Enhanced MR Images from acquisition having references to the PR
- Graphics for Enhanced MR Images requires a PR!
  - No other graphics allowed (private graphics, Overlay Bitmap)!
- Multiple PR possible.

### Presentation State

- label/annotation
- windowing
- zoom
- panning
- rotations



## 10 Interoperability Export & Formats

New for Numaris/X: the export format needs to be defined for each node.  
To validate the settings Dicom Testimages are available (see 0 for details)  
The following gives an high level overview:

### Syngo.via.

- Enhanced DICOM supported

### Syngo

- Enhanced DICOM support for: DICOM, Patient Browser, Viewer, 3D, Filming
- No Enhanced DICOM: MR Applications, Exam

### PACS

- Enhanced DICOM support is vendor dependent
- Aspects to be checked: archiving, 2D, 3D, PR support

### DICOM send

- Format setting defined per DICOM node in Administration Portal
- Presets from Clinical IT/Customer → fixed setup for Clinical user

### Media Export

- Preconfigured profiles in Configuration Panel
- User chooses profile

### File Export

- User controls format

## 11 History

| Rev. | Date       | Product / Version                                                                                                                                  | Author         | Change and reason for change                |
|------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------------------------------------------|
| 1.0  | 31.06.2018 | MAGNETOM Vida<br>MAGNETOM Sola /<br>NUMARIS/X VA11A                                                                                                | Markus Zetsche | Adapted for Software<br>Version VA11A       |
| 2.0  | 02.04.2019 | MAGNETOM Vida<br>MAGNETOM Sola<br>MAGNETOM Lumina<br>MAGNETOM Altea /<br>NUMARIS/X VA11B<br>MAGNETOM Amira<br>MAGNETOM Sempra<br>/ NUMARIS/X VA12M | Klaus Gartner  | Adapted for Software<br>Version VA11B/VA12M |
|      |            |                                                                                                                                                    |                |                                             |
|      |            |                                                                                                                                                    |                |                                             |
|      |            |                                                                                                                                                    |                |                                             |

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